# nixlab – Project documentation

As of: 2026-10-19

> This documentation describes a private repo. The domain and internal addresses are replaced with example values: the domain with `homelab.example` (RFC 2606), IPv4 per RFC 5737 (`192.0.2.0/24`, `198.51.100.0/24`, `203.0.113.0/24`), IPv6 per RFC 3849 (`2001:db8::/32`), the tailnet name with `example-tailnet.ts.net`. The real mesh range is a /22 out of RFC 1918. Modules, documentation and build tooling were written together with Claude (Anthropic). License: MIT.

## Overview

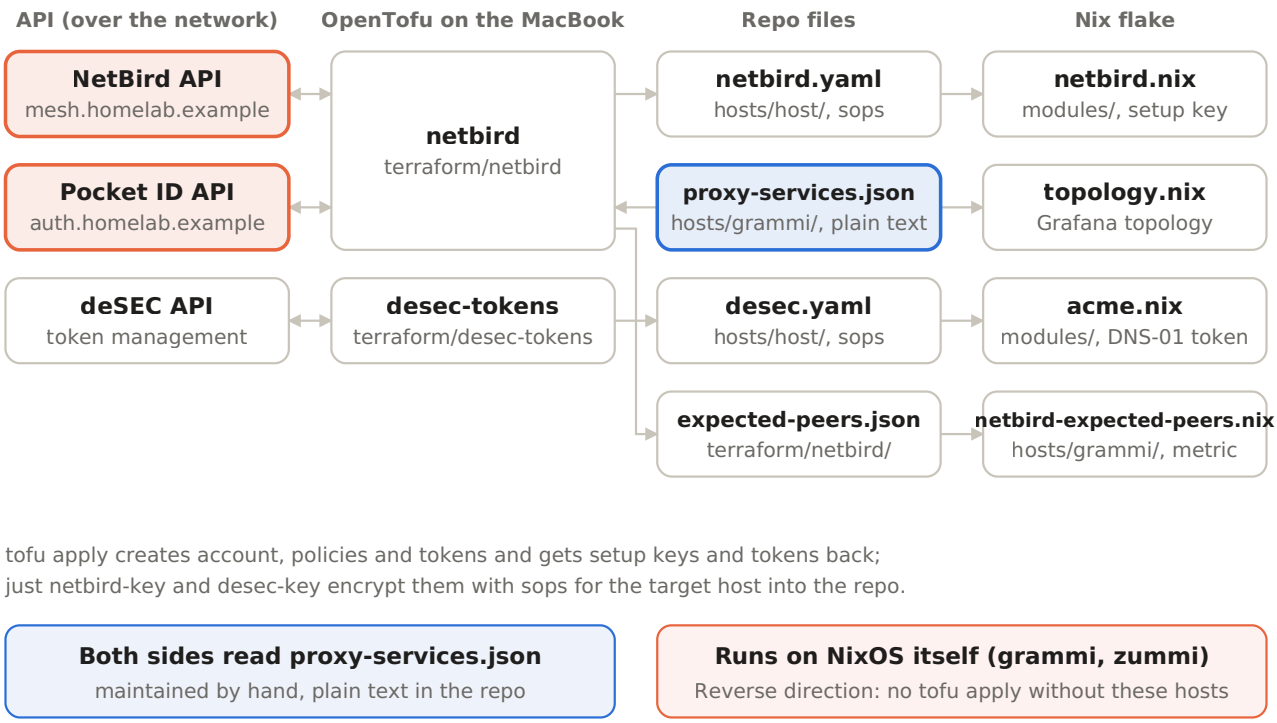
nixlab is the flake monorepo for the **homelab.example** homelab: five NixOS hosts, a Mac mini running nix-darwin and the admin workstation gusto (a MacBook, nix-darwin + home-manager), plus three OpenTofu modules for the NetBird account, the deSEC API tokens and the GitHub repositories with their deploy keys. Everything is operated through a `justfile` from gusto; the hosts pull their own configuration from the private GitHub repo and update themselves overnight.

Short answer to the question "*Did you wire NixOS and OpenTofu together somehow?*": **Yes, but deliberately loosely – through files in the repo, not through a shared tool.** OpenTofu writes sops-encrypted files into `hosts/<host>/` that the NixOS modules read at build time, and both sides read the same service list `hosts/grammi/proxy-services.json`. No terranix, no Tofu run from within Nix. Details in NixOS & OpenTofu.

Audience: people who can read NixOS flakes, sops-nix and Terraform/OpenTofu.

## NixOS & OpenTofu

Yes, the two are wired together – with the repo as the shared interface. OpenTofu manages what exists outside the machines behind an API (NetBird account, deSEC tokens, Pocket ID groups); NixOS manages the machines. The two never talk to each other directly: Tofu only runs on the MacBook through `just`, Nix never evaluates Tofu state.




Tofu never writes to a host directly: every result ends up as a file in the repo and reaches the host only with the next deploy.

## The touch points

| Direction | File | Created by | Read by | Contents |
|---|---|---|---|---|
| Tofu → NixOS | `hosts/<host>/netbird.yaml` | `just netbird-key <host>` (`tofu output setup_keys` → `sops encrypt`) | `modules/netbird.nix` (`sopsFile`) | One-time setup key, valid for 24 h |
| Tofu → NixOS | `hosts/<host>/desec.yaml` | `just desec-key <host>` (`tofu output acme_tokens` → `sops encrypt`) | `modules/acme.nix` (`sopsFile`) | deSEC token for TXT records only, VMs pinned to their IP |
| Tofu → NixOS | `terraform/netbird/expected-peers.json` | `just netbird-apply` (`tofu output expected_peers`, computed from the policies in `expected-peers.tf`) | `hosts/grammi/netbird-expected-peers.nix` (`builtins.readFile`) | Expected NetBird peers per host as metric `netbird_expected_peers`, next to the real network map in the NetBird dashboard (e.g. aldrin 6, grammi 8) |

| Direction | File | Created by | Read by | Contents |
|---|---|---|---|---|
| Tofu → NixOS | `terraform/netbird/admin-peers.json` | `just netbird-apply` (`tofu output admin_peers`, from the group `admin-workstations`, `admin-peers.tf`) | `flake.nix` (`builtins.readFile`, option `services.docsSite.allow`) | Mesh addresses of the admins' devices – who may read the private docs |
| shared | `hosts/grammi/proxy-services.json` | by hand | `terraform/netbird/reverse-proxy.tf` (`jsondecode(file(...))`) and `topology.nix` | Name → app, port, groups of the reverse proxy services |
| repo → both | `hosts/<host>/github.yaml` + `github-deploy.pub` | `just github-key <host>` (`ssh-keygen` → `sops encrypt`) | private half: `modules/deploy.nix` (`sopsFile`); public half: `terraform/github` (`file(...)`) | Deploy key for `/etc/nixos`; Tofu only ever sees the public half |
| NixOS → Tofu | NetBird server on grammi | `hosts/grammi/netbird.nix` | provider via `https://mesh.homelab.example` | The API Tofu runs against in the first place |
| NixOS → Tofu | Bootstrap tokens | `just netbird-setup-owner`, `netbird-proxy-token` (SSH to grammi, server API) | `terraform/netbird/secrets.yaml` and `hosts/grammi/netbird-proxy.yaml` | Owner PAT, proxy token |

The flow for a new host: Tofu creates the setup key, `just netbird-key` writes it encrypted for exactly that host (age key derived from its SSH host key) to `hosts/<host>/netbird.yaml`, commit, deploy within 24 h. The host registers itself on start; Tofu does not manage peers, it looks them up by name via `data "netbird_peer"`.

### What is deliberately not wired

- **No shared host list.** Hosts are listed three times: `nixosConfigurations` in `flake.nix`, `local.hosts` in the Tofu modules and `hosts :=` in the `justfile`. A new host has to be added in all three places.
- **No terranix / no Tofu in the Nix build.** Tofu state is local (gitignored, in the main checkout, also from worktrees) because it holds setup keys and tokens in plain text.
- **No provisioning of the machines.** The VMs are created by hand at the provider, ursalia/cuckoosegg following `docs/install-*.md`.
- **Public DNS** lives in DNSControl (its own repo, only linked in via symlink), not in Tofu.

So the line is: whatever Nix can build deterministically from the repo, Nix does. Whatever is created once through an API and returns a secret, Tofu does – and hands it over encrypted as a file.

## Secrets, network & access

### Secrets (sops-nix, age)

Every host file is encrypted for exactly two recipients: the age key derived from the host's SSH host key and the offline recovery key `admin`. So the MacBook cannot read host secrets – `just secrets <host>` edits them on the host and fetches the result back. Encrypting for a host, on the other hand, works anywhere (only the public key is needed); that is exactly what `netbird-key` and `desec-key` use.

| File | Recipients | Contents |
|---|---|---|
| `hosts/<host>/secrets.yaml` | host + admin | Deploy key, app passwords, OIDC clients, restic |
| `hosts/<host>/netbird.yaml` | host + admin | NetBird setup key (from Tofu) |
| `hosts/<host>/desec.yaml` | host + admin | deSEC token (from Tofu) |
| `hosts/<host>/github.yaml` | host + admin | GitHub deploy key (from `just github-key`) |
| `terraform/*/secrets.yaml` | workstation + admin | API tokens for the providers |

The workstation key lives in 1Password and is read via `SOPS_AGE_KEY_CMD`, never as a file. `env.sh` in each Tofu module decrypts the tokens only into the shell of the respective recipe.

## Network

- **NetBird mesh** (`192.0.2.0/24`, peer names under `peer.homelab.example`): control plane on grammi, login via Pocket ID. Private range (RFC 1918) instead of CGNAT, because tailscaled drops packets from 100.64/10 while both clients run in parallel.
- **Groups:** `servers` (all hosts except cuckoosegg), `subraum-devices`, `admin-workstations` (MacBook), plus person groups from Pocket ID.
- **Home LAN** (`198.51.100.0/24`): routed through ursalia (metric 100), with the two Apple TVs as fallback (group `home-routers`, metric 200, own setup key via `just netbird-router-key`, owned by nobody, in no policy); for group `home-lan`, not for devices that are already at home (posture check `not-at-home`). The Apple TVs should have "Quantum-Resistance" on in the app; since 2026-10-21 the mesh runs Rosenpass permissive, so it works without, just not quantum-resistant.
- **ursalia's uplink:** WLAN into the home LAN's main network (Intel BE200, `hosts/ursalia/wifi.nix`): wpa_supplicant, WPA3 with WPA2 fallback, country DE, all bands, passphrase via sops. Its final spot has no cable; eno1 keeps DHCP as a fallback with a worse route metric, so the WLAN stays the primary link when both are up.
- **zorlock's uplink:** WLAN as well (`en1`, Broadcom); `hosts/zorlock/wifi.nix` makes sure `homelab` is a preferred network (`networksetup`, only ever adds it, passphrase via sops in `hosts/zorlock/wifi.yaml`). nix-darwin has no Wi-Fi module; band choice stays with macOS.
- **Subraum:** routed through cuckoosegg for `subraum-access`. cuckoosegg itself may only reach grammi's ingest (TCP 8428/9428) and may only be reached from the MacBook via SSH.
- **Tailscale** still runs in parallel during the migration and serves as break-glass: `NIXLAB_DOMAIN=example-tailnet.ts.net just deploy zummi`.

## Access

Only 22/80/443 are open to the public. `modules/ssh.nix` allows root SSH only from mesh or Tailscale addresses; every host has its own login keypair. The mesh is IPv4 only for now; dual stack is listed as an open item.

### Private docs: admin devices only

The private version of this documentation (`doku.homelab.example`, on aldrin and ursalia) contains every internal address and describes how the lab is run. That is more than every device in the mesh needs to see. So nginx serves it only to the mesh addresses of the admins' devices; everyone else gets 403 – with a page of its own that says why, which address the request came from and how a device gets in (without the navigation, so the page titles stay hidden). The name itself resolves for all peers.

**From outside** the same name leads to the public, anonymised docs (GitHub Pages): publicly `doku` is a CNAME to the reverse proxy, whose service `doku` hands it on without a login to a listener on aldrin (port 8091, mesh only) that knows nothing but a redirect. Whoever is in the mesh gets to aldrin and ursalia directly through the NetBird zone.

| Who | Path | Result |
|---|---|---|
| Outside, or NetBird off | reverse proxy (grammi) → aldrin:8091 | 301 to the public docs |
| In the mesh, device in `admin-workstations` | straight to aldrin/ursalia | private docs |
| In the mesh, any other device | straight to aldrin/ursalia | 403 page |

**Why through the proxy:** this way `doku` is listed with the other public names in the NetBird dashboard instead of being a public vhost of its own on aldrin. The proxy cannot redirect by itself, only forward – hence the redirect listener. It has a port of its own rather than the docs vhost recognising the proxy by its address: the proxy is a peer of its own whose mesh address Tofu does not know. Linking `homelab.example` to the public docs is accepted on purpose; the real addresses stay replaced there.

**What counts as an admin device:** the devices in the group `admin-workstations` (`terraform/netbird/subraum.tf`), today the MacBook – the same device that may also SSH to the Subraum devices. Deliberately not the person group `homelab_admins`: that one is filled from Pocket ID with *all* devices of an admin user. Today that would be none (Calla has no devices of its own), and a phone signing in under an admin user would be in at once. An admin here is a device named in Tofu, not a login. A further device in `admin-workstations` therefore also gets SSH to the Subraum devices – the same trust. Tofu reads the members on every `just netbird-apply` and writes name → address to `terraform/netbird/admin-peers.json`; `flake.nix` hands the addresses to `services.docsSite.allow`. A new admin device therefore gets in after `just netbird-apply`, a commit and a deploy of aldrin and ursalia.

**Why nginx and not a NetBird policy:** `to-servers` lets the devices of all persons (`homelab` and `homelab_admins`) reach all ports on all servers. NetBird policies can only allow, never deny – a narrower rule for the docs would change nothing as long as `to-servers` stays as it is. Narrowing `to-servers` to a port list per service would work, but touches every service on every server for one static site. Moving the docs to a port of their own fails at the same point. So the policy still decides which hosts a device reaches, and nginx decides who gets the docs.

**Why no login (Pocket ID):** you read the docs when something is broken. A login would tie them to zummi (Pocket ID) and, through the reverse proxy, to grammi – exactly the hosts they are meant to be independent of. An address list needs nothing at runtime but nginx on the host itself.

**What this deliberately accepts:**

- The list is as current as the last `just netbird-apply` followed by a deploy. Until then a new admin device sees 403, and a removed device still gets in.
- The address stands for the device, not the person: whoever uses an admin device reads the docs.
- The list must never be empty – `just netbird-admin-peers` refuses an empty output, and the module fails the build with an assertion instead of quietly giving everyone 403.
- IPv4 only, like the mesh. With the IPv6 overlay the v6 addresses go into the same list.

**Paperless: from outside, share links (PIN) and the app (SSO)**

In the mesh `dokumente.homelab.example` resolves straight to ursalia through the NetBird zone (full app, Pocket ID login).

**From outside** `dokumente` is a CNAME to the reverse proxy. Its service `dokumente` offers two logins and hands both on to a listener on ursalia (port 8092, mesh only, HTTP through the tunnel):

- **PIN**, for third parties with a share link: `share/<slug>` only (documents and bundles, both paperless' `SharedLinkView`), 404 otherwise. An expired or unknown link makes paperless redirect to its login page; the listener answers that with a short note.
- **SSO** via Pocket ID (groups `paperless-users`/`paperless-admins`): the full app including paperless-gpt, as in the mesh. Paperless' own Pocket ID login still follows, usually without a prompt.

| Who | Path | Result |
|---|---|---|
| Outside with PIN | reverse proxy (grammi) → ursalia:8092 | `/share/<slug>` only, 404 otherwise |
| Outside with SSO | reverse proxy (grammi) → ursalia:8092 | full app |
| In the mesh | straight to ursalia:443 | full app, share links without PIN |

**How the listener recognises SSO:** by the header `X-NetBird-User`. The proxy deletes whatever the client sends under that name on every request and sets it to the email address for SSO sessions only (`stampNetBirdIdentity` in NetBird, checked with 0.81.0 – check again on NetBird updates). The header only picks the paths, it logs nobody in: paperless still asks for its own login. A mesh peer sending the header straight to port 8092 gets nothing it would not get on 443 anyway.

**Why the same name:** paperless builds a share link from the origin of the UI. A link `https://dokumente.homelab.example/share/<slug>` therefore works for third parties without rewriting – whether created in the mesh or via SSO. A name of its own for either login would mean editing links by hand.

**Why the listener filters, not the PIN:** the PIN is a second hurdle in front of the unguessable slug, not a replacement for it. That PIN visitors do not see the app rests on nginx on ursalia alone. The proxy service has no path filter (target `path`), otherwise the redirect to `/accounts/login/` for expired links would no longer reach the listener.

**PIN:** exactly 6 digits (the NetBird API accepts nothing else), one for all shares. New one: `just dokumente-share-pin` (random, into `terraform/netbird/secrets.yaml`), then `just netbird-apply`; show it: `just dokumente-share-pin-show`. The proxy blocks a client address for 15 minutes after 5 wrong PINs within 5 minutes.

**What this deliberately accepts:**

- A still-valid PIN session (the proxy's cookie) skips the login page. Whoever then wants the app via SSO lands on a note page and has to clear the site data or use a private window first.
- Whoever knows the PIN gets every link whose slug they know – no others. So create shares in paperless with an expiry date.
- Telling PIN from SSO rests on a behaviour of the proxy, not on a documented interface.
- CrowdSec sees neither PIN attempts nor requests to the app (TCP passthrough, no HTTP log on grammi), as with every proxy name; the proxy's own block and paperless' login remain.
- ursalia's nginx logs the proxy's mesh address as the client, not the visitor's (no `realip`: it would rewrite the address before `allow`/`deny`).

**MQTT from outside: TLS passthrough to Home Assistant**

`mqtt.homelab.example:8883` is the Mosquitto add-on on Home Assistant, for MQTT devices that are also used away from home (AWTRIX NG on an Ulanzi TC002). Same name, port and certificate everywhere, so a device is configured once:

| Where | Path |
|---|---|
| At home | the home router (OPNsense) answers `mqtt.homelab.example` with HA's LAN address (local DNS entry) → HA:8883 directly, also without internet |
| Outside | public CNAME to the reverse proxy → `netbird-proxy` on grammi, port 8883 → through the mesh to HA:8883 |

**HA is a NetBird peer** (NetBird add-on, `terraform/netbird/home-assistant.tf`), in a group of its own `home-assistant` – not in `servers`, so it reaches exactly two things and nothing reaches it but the proxy. NetBird's DNS management is off for that group: HA keeps resolving through the home router (OPNsense), also when the internet is gone.

**TLS ends in Mosquitto, not at the proxy.** The service `mqtt` (`terraform/netbird/reverse-proxy.tf`) runs in L4 mode `tls` with the peer HA as its target: the proxy reads only the SNI and passes the connection through untouched. For an L4 service with a port of its own the proxy binds that port itself on all addresses (only 127.0.0.1 has to send a PROXY header there); grammi's firewall opens 8883 in `hosts/grammi/netbird-proxy.nix`.

**The certificate** covers `homeassistant.homelab.example` and `mqtt.homelab.example`. HA's Let's Encrypt add-on gets it via DNS-01 – but without a deSEC token of its own (it could not be pinned to an address, the home address changes). It asks dns01proxy on grammi instead (`hosts/grammi/dns01proxy.nix`, provider `dns-httpreq`), at `https://acme.homelab.example:8446`:

- The port is open only on the mesh, and the policy `home-assistant-acme` lets only HA reach it. HA resolves the name through the home router (OPNsense) (local entry to grammi's mesh address, `tofu output acme_proxy_address`), every other peer through the NetBird zone.
- The proxy writes the TXT records with grammi's address-pinned token. HA's account (password in the add-on, bcrypt hash in `hosts/grammi/dns01proxy.yaml`, `just dns01proxy-key-homeassistant`) may only answer challenges for the two names above.
- dns01proxy serves HTTPS itself and fetches its own certificate for `acme.homelab.example` the same way.

**The two local DNS entries** are host overrides in Unbound on the OPNsense (Services → Unbound DNS → Overrides), set by hand – the API key `inventory-ro` is read-only:

| Name | Address | Why |
|---|---|---|
| `mqtt.homelab.example` | 198.51.100.168 (HA's LAN address) | MQTT stays in the home network and works without internet |
| `acme.homelab.example` | 192.0.2.122 (grammi's mesh address, `tofu output acme_proxy_address`) | HA reaches dns01proxy; NetBird does not manage HA's DNS |

If HA's address or grammi's mesh address changes (new lease, NetBird re-registration), the override has to follow by hand.

The add-on does not renew by itself: an HA automation starts it daily and restarts Mosquitto weekly, so the renewed certificate gets loaded.

**What this deliberately accepts:**

- No login at the proxy – L4 has none. Mosquitto's own users are the only gate, and that includes every login the add-on accepts (its own list and Home Assistant users).
- Mosquitto sees the proxy's mesh address as the sender of every outside connection (the add-on cannot take a PROXY header), so neither HA nor CrowdSec can block a single client by address. The proxy can filter by country or address range (`access_restrictions`), not set for now.
- Certificate renewal needs grammi and the mesh – with 30 days of slack before expiry.
- A device that restarts while the internet is down needs the time from somewhere other than the internet to check the certificate.

## Deployment & operations

The pull hosts (zummi, aldrin, grammi, ursalia) keep `/etc/nixos` as a checkout of the private GitHub repo and build themselves. cuckoosegg is pushed to, zorlock and gusto are switched on the Mac itself with `darwin-rebuild` (macOS hosts).

| Host | Manual deploy | Automatic |
|---|---|---|
| zummi, aldrin, grammi, ursalia | `just ship <host> "<message>"` = commit, push, `git pull` + `nixos-rebuild switch` on the host | nightly auto-upgrade, reboot window 04:00–06:00 |
| cuckoosegg | `just deploy-push cuckoosegg` (built on ursalia, copied to the target) | ursalia pushes daily at 06:30 |
| zorlock | `just mac-switch zorlock` (= `deploy-zorlock`: `git pull` + `darwin-rebuild switch` with sudo, with a summary) | none |
| gusto | `just mac-switch` on gusto, from the working copy | none – `just mac-upgrade` for the Homebrew pins |

### Nightly upgrade

`modules/auto-upgrade-git.nix` updates `nixpkgs`, `nixpkgs-unstable` and `sops-nix`, builds, switches and pushes the new `flake.lock` back to GitHub with the host's deploy key. The GitHub host key is pinned because nobody is watching. A configuration error that only shows up on a cold boot therefore takes the host down at 04:00 – for the VMs there is the provider's console for that.

## Monitoring

Everything ends up on grammi: **vmagent** on every host sends metrics to VictoriaMetrics (`:8428`), **Vector** sends journald and nginx logs to VictoriaLogs (`:9428`), both over the mesh. Grafana (`metrics`) and logchef (`logs`) sit behind the reverse proxy with SSO. Exporters per host come from `modules/metrics.nix`; whatever has no endpoint of its own goes through node_exporter's textfile collector (`/var/lib/prometheus-node-exporter-text-files`), written by small timers or services:

| Textfile metric | Host | Source |
|---|---|---|
| `tailscale_*` | hosts with Tailscale | `tailscale metrics write`, every 30 s |
| `fwupd_updates_*` | ursalia | `fwupdmgr get-updates` after each metadata refresh |
| `wifi_link_*` | ursalia | `iw dev wlp2s0f0 link` (PHY mode, width, MCS, NSS), every 30 s |
| `node_wifi_*`, `wifi_link_*` | zorlock | `system_profiler SPAirPortDataType -json` every 60 s, same names as on ursalia (no receive rate, retries or separate receive PHY mode on macOS; macOS redacts SSID/BSSID without Location Services even for root, shown as "n/a" in the dashboard), plus `wifi_link_noise_dbm` |
| `netbird_expected_peers` | grammi | `terraform/netbird/expected-peers.json`, baked into the system at build time |

The dashboards are JSON in `hosts/grammi/dashboards/`, provisioned from the repo (changes only via commit + deploy of grammi). `topology.json` is generated by `just topology`.

| Dashboard | Shows |
|---|---|
| Hosts Overview | Scrape targets, CPU/memory/disk/network per host, failed units, certificates, backups (btrbk, restic), pending firmware, WLAN band/signal |
| Topology | Hosts, exporters and data flows as a Graphviz diagram |
| NetBird | Control plane reachable per host; peers per host as expected (Tofu) / known (network map) / connected P2P / connected relay, plus deviations; connection history, P2P latency, sync and setup times; netbird-server (signal, relay, gRPC, API) |
| WLAN | ursalia's and zorlock's band (2.4 / 5 / 6 GHz) over time, channel, signal, PHY mode, link rate, traffic, MCS, retries, time associated |

| Dashboard | Shows |
|---|---|
| Disk Health | SMART/NVMe of ursalia: status, critical warnings, wear, spare, temperature, data written |
| Paperless | Health (Celery, Redis, DB, index, classifier), documents, inbox, failed tasks, storage, log warnings |
| paperless-gpt | OCR and suggestion backlog, OCR time per page, LLM requests, durations, tokens, errors; zorlock's GPU and memory |
| Web & Security | nginx and Traefik requests, status codes, durations; CrowdSec bans, alerts, parser hits |
| Garage (S3 storage) | aldrin's Garage: cluster health, disk, S3 requests/errors/duration, resync and GC queues |
| VictoriaMetrics & VictoriaLogs | Ingest rate, disk usage, query duration, active series |
| zorlock (Mac mini) | mactop: thermal state, CPU/GPU, power, temperatures, memory, fan; Ollama requests, speed, model load times |

**Expected vs. known peers (NetBird):** "expected" is computed by OpenTofu from the policies with the group membership at the last `just netbird-apply` (example: aldrin 6, grammi 8, cuckoosegg 4). "known" is what the management server really distributes. A device that joins later raises known at once, expected only after the next apply + deploy of grammi – on purpose, so a new peer is visible. The model ignores posture checks; see the header of `expected-peers.tf`.

**Grafana pitfall:** state timelines with `mergeValues` and colour mode "thresholds" turn numbers into threshold buckets ("1+", "< 1") before value mappings are applied. Use colour mode "fixed" and put text and colour into the mappings.

## macOS hosts

Two machines in this repo run macOS with nix-darwin instead of NixOS: **zorlock**, the Mac mini serving Ollama, and **gusto**, the MacBook that is the admin workstation. Both are `darwinConfigurations` in the same flake and use the same darwin-tested nixpkgs (`nixpkgs-darwin`). Apart from that they are opposites – one is a server nobody sits at, the other a laptop that is only ever used by a person – and several things are therefore solved differently on purpose. This page explains what the two share, where they differ, and why.

| | zorlock | gusto |
|---|---|---|
| Role | LLM server (Ollama for paperless-gpt) | Admin workstation: deploys, secrets, Tofu, docs |
| Configuration | `hosts/zorlock/` | `hosts/gusto/` (system) + `home.nix`, `ssh.nix` (home-manager) |
| Built and switched | on zorlock, from `/etc/nixos` (`just mac-switch zorlock`) | on gusto, from the working copy (`just mac-switch`) |
| Homebrew | off | on – the apps are casks |
| home-manager | no | yes, for admin's dotfiles |
| NetBird client | nix-darwin `services.netbird`, client from `nixpkgs-netbird` | Homebrew cask `netbird-ui` from NetBird's tap |
| NetBird login | setup key from sops, unattended | SSO (Pocket ID) in the app |
| SSH into it | yes, as admin, keys only | no – Remote Login stays off |
| sops secrets | yes (deploy key, deSEC token, NetBird key, Wi-Fi) | none |
| Monitoring, logs | vmagent, Vector → grammi | none – not in the topology |

| | zorlock | gusto |
|---|---|---|
| Availability | always on (`criticality: service`) | intermittent (`kind: laptop`) |

None of `modules/*.nix` applies to either Mac – those modules are written for NixOS (systemd, ifstate, nginx). What does carry over is restated in each host's `configuration.nix` in nix-darwin's terms. The one module the two share is `modules/darwin/homebrew.nix`.

## Homebrew: one policy for both Macs

Some macOS software exists only as a Homebrew cask, or should come signed by its vendor: 1Password, Vivaldi, Arq, Ghostty, the NetBird app. `modules/darwin/homebrew.nix` is imported by every `darwinConfiguration` and sets one policy (decided on 2026-10-21). It only takes effect where a host sets `homebrew.enable = true` – today that is gusto; on zorlock the module changes nothing, its system closure is identical with and without it.

**Homebrew and its taps are pinned in `flake.lock`.** Homebrew itself is not installed with the `curl | bash` script but by the `nix-homebrew` input. `homebrew-core`, `homebrew-cask` and every third-party tap are flake inputs (`flake = false`), with `mutableTaps = false`. There is no git checkout for `brew update` to move: which version of a cask a switch installs depends on `flake.lock` alone – the same logic as for nixpkgs. A host adds a tap with one line,

```
nix-homebrew.taps."netbirdio/homebrew-tap" = inputs.netbirdio-tap;
```

and `homebrew.taps` (the list nix-darwin writes into the Brewfile) is derived from that set automatically.

**A switch does what the configuration says – and nothing else.**

- `onActivation.autoUpdate = false`: no `brew update` during a switch.
- `onActivation.upgrade = false`: no `brew upgrade` during a switch either; an installed cask stays at its version until it is upgraded on purpose.
- `onActivation.cleanup = "zap"`: whatever is installed but not declared is removed, casks including their application data (`brew uninstall --zap`). Deleting a line from `homebrew.casks` therefore really uninstalls the app with its settings. `just mac-diff` shows this before it happens.
- `global.autoUpdate = false` and `global.brewfile = true`: the same for `brew` typed by hand, and `HOMEBREW_BUNDLE_FILE` points at the generated Brewfile, so a plain `brew bundle check` compares against the configuration.

Apps with their own updater, and Mac App Store apps, keep updating themselves; the policy only concerns what Homebrew does.

**No `brew shellenv`.** Neither nix-darwin's nor nix-homebrew's shell integration is used – both would run `brew shellenv` in every new shell and put Homebrew *in front of* Nix in `PATH`. Instead `/opt/homebrew/bin` and `sbin` are added to the system `PATH` directly, after the Nix profiles and before macOS' own directories:

```
~/.nix-profile/bin : /etc/profiles/per-user/$USER/bin : /run/current-system/sw/bin : /nix/var/nix/
profiles/default/bin : /opt/homebrew/bin : /opt/homebrew/sbin : /usr/local/bin : /usr/bin : …
```

A tool from Nix always wins over the same tool from Homebrew, Homebrew wins over Apple's stubs, and opening a shell starts no `brew` at all.

### The mac-* recipes

| Recipe | What it does |
|---|---|
| `just mac-diff [host]` | Builds the configuration and shows what a switch would change: the Nix closure (`nvd diff`), the casks and formulae that would be **removed** (`brew bundle cleanup --zap` without `--force`) and those that would be **installed**. Changes nothing. |
| `just mac-switch [host]` | Builds, switches, and ends with a compact summary: Nix packages added/removed/updated (nvd) and Homebrew `- removed` / `+ new`. |
| `just mac-upgrade` | Updates only the Homebrew pins (`nix-homebrew`, `homebrew-core`, `homebrew-cask`, `netbirdio-tap`), switches, runs `brew upgrade` and prints `old -> new` per upgraded item, plus the NetBird client version on gusto next to the mesh's pin. The result is a `flake.lock` change to commit. |

The host defaults to `gusto`, which is built from the working copy (uncommitted changes included – but nix only sees files git knows, so `git add` new files first). `zorlock` is reached over SSH: `git pull` in `/etc/nixos`, then the same script under sudo. `deploy-zorlock` and `check-zorlock` are now short forms of `mac-switch zorlock` and `mac-diff zorlock`. The work itself is done by `modules/darwin/rebuild.sh` on the Mac.

`mac-upgrade` exists only for gusto: it changes `flake.lock`, which is committed on the workstation; zorlock has no Homebrew.

## NetBird on the two Macs

Both Macs are peers in the NetBird mesh, but they get their client in two different ways. This is deliberate.

### zorlock: nix-darwin module, client from the mesh pin

`hosts/zorlock/netbird.nix` turns on nix-darwin's `services.netbird`. That module runs only the daemon – `netbird service run` as a launchd daemon, as root, state in `/var/lib/netbird` – and leaves login and settings to the user. The rest is done by a second launchd job, `netbird-login`, the same pattern as on the Linux hosts:

- The **client binary** comes from the `nixpkgs-netbird` input, the pin the NetBird server on grammi and every NixOS client use. The whole mesh moves together with `nix flake update nixpkgs-netbird` – server first, then the clients.
- **Login** is unattended: `netbird-login` registers once with a single-use setup key from `hosts/zorlock/netbird.yaml` (sops, `just netbird-key zorlock`). After that the key is spent and the peer's identity lives in `/var/lib/netbird/config.json`.
- **Rosenpass** (permissive, like everywhere) and the client's local Prometheus endpoint (`127.0.0.1:9191`, scraped by vmagent) are turned on by the same job – nix-darwin's module has no `config.d` like the NixOS one.
- After a reboot zorlock joins the mesh only once FileVault has been unlocked (over SSH from the home LAN), because the daemon lives on the data volume.

That is what a server needs: no GUI session (zorlock should stay logged out, Ollama counts free RAM), coming back by itself, the same client version as the rest of the mesh, and metrics.

### gusto: the official app as Homebrew cask

gusto installs `netbirdio/homebrew-tap/netbird-ui`, the cask from NetBird's own tap, pinned as the flake input `netbirdio-tap` (decided on 2026-10-21). The cask is the signed menu-bar app; it depends on the formula `netbird` from the same tap (the CLI and daemon), and its installer registers the daemon as `/Library/LaunchDaemons/netbird.plist`. That is the same software the official `.pkg` installer puts on a Mac,

only installed and removed through the configuration. The formula is also listed on its own (`homebrew.brews`), so `cleanup = "zap"` can never treat the daemon as an orphan and remove it from under the app.

- **Login** is interactive: in the app, with SSO through Pocket ID – the peer then belongs to the admin's user and its groups, the way a person's device should. There is no setup key and no sops file for gusto.
- **Rosenpass** is set once in the app's settings (Quantum-Resistance); the daemon keeps it in its own config. Since 2026-10-21 it is **off** on the workstation – a test against tunnels that stayed "Connected" without a WireGuard handshake after the Rosenpass key expired (upstream #3107, #3121); every other peer stays permissive so it still accepts the workstation. gusto starts the same way, off. If the dead tunnels persist, Rosenpass goes off everywhere; if they are gone, the setting stays as it is until upstream fixes the re-keying.
- **Version**: whatever the pinned tap holds. It changes only with `just mac-upgrade`, which prints gusto's client version next to `nixpkgs-netbird`'s so the two do not drift far apart. The usual rule still applies: the server (grammi) is updated first, clients follow.
- **Updates from elsewhere**: the app's own update notice and NetBird's dashboard-driven auto-update must not move the version behind the configuration's back; this is checked on the first switch and switched off if the app offers it.
- No metrics endpoint and no log shipping – gusto is not monitored.

#### Why not the same on both?

| Question | Answer |
|---|---|
| Why does gusto not use the nix-darwin module like zorlock? | That module has no menu-bar app; everything would be CLI. On a laptop that changes networks all the time, the app is how one sees at a glance whether the mesh is up, switches profiles and signs in again. Login on gusto is SSO anyway, which needs a browser and a person – the unattended setup-key job that makes zorlock's variant worthwhile does not apply. |
| Why does zorlock not use the cask? | A server must come back on its own after a reboot, without anybody signing in, with the same client version as the mesh and with metrics. The cask's app needs a GUI session, and its version follows the tap, not the mesh pin. |
| Why not `netbird-ui` from nixpkgs on gusto? | nixpkgs packages the UI for Linux first; whether it builds and runs properly on macOS (signing, the menu-bar integration) is not verified, and it would not be the signed binary NetBird ships. |
| Mac App Store? | NetBird offers no Mac App Store version (only iOS). |
| What is the price? | gusto's client version is pinned separately from the mesh's (`netbirdio-tap` instead of `nixpkgs-netbird`). `just mac-upgrade` makes the gap visible; it is closed by hand. |

#### The peer in Tofu

`terraform/netbird/subraum.tf` looks up the admin workstation by peer name (`data "netbird_peer"`) for the group `admin-workstations` – the group that may reach the Subraum devices and the docs site. Changing the workstation (old MacBook → gusto) is done with both peers in the group for a short while: add a second data source `gusto`, `just netbird-apply`, deploy aldrin and ursalia (docs site allow-list), test, and only then remove the old peer.

## gusto in detail

The reasons behind gusto's setup; the complete list of what is on it, every setting and the day-to-day tasks are on gusto – the admin workstation.

### Starting from scratch

gusto was set up deliberately bare (2026-10-21): no app list taken over from the old installation, only a small base. Everything else is added the moment it is needed – a line in `hosts/gusto/configuration.nix` (cask) or `home.nix` (CLI tool), `just mac-switch`, and, if there is data, it comes back from the backup.

| Where | What |
|---|---|
| Casks | 1Password + `1password-cli`, Vivaldi, Arq, Claude, Ghostty, NetBird |
| System (`environment.systemPackages`) | git, gh, just, jq, neovim (`EDITOR`), curl, wget, rsync |
| admin (home-manager) | zsh, atuin, fzf, zoxide, direnv + nix-direnv, bat, eza, ripgrep, fd, tree, lazygit, btop, yazi, tmux, nvd, nix-output-monitor, uv, typst |
| Font | Maple Mono NF (Nerd Font) for Ghostty and the TUIs |
| Repo tools | sops, age, ssh-to-age, opentofu, nvd – from the flake's devShell; dnscontrol from a separate one for the DNSControl repo, see below |

`op` comes as a cask and not from nixpkgs because the CLI unlocks through the 1Password app with Touch ID, which only works with the vendor's signed binary.

### XDG and home-manager

Project rule: configuration lives under `~/.config/<tool>` wherever the tool supports it – zsh (`ZDOTDIR=~/.config/zsh`), git (`~/.config/git/config`), Ghostty, tmux, direnv. A tool without XDG support keeps its real file in the classic place and gets a link `~/.config/<tool>` pointing at it, e.g. `~/.config/ssh → ~/.ssh`. Data goes to `~/.local/share`, state (zsh history) to `~/.local/state`.

`PATH` gets exactly one personal directory, `~/.local/bin`: `GOBIN` points there, `CARGO_INSTALL_ROOT` is `~/.local`, and `uv tool install` uses it by default. `GOPATH` and `CARGO_HOME` live under `~/.local/share`. No `~/go/bin`, `~/.cargo/bin` or pnpm path any more.

git: name `admin`, default address `mail@admin.org`; for everything below `~/Projects/homelab/` an `includeIf gitdir:` switches to `admin@homelab.example`.

### SSH: 1Password only

The private keys exist only in 1Password; its agent offers them. Two links make the agent reachable under a path without spaces: `~/.1password/agent.sock` (home-manager) and `SSH_AUTH_SOCK` – in every shell (home-manager) and for GUI apps through `launchd.user.envVariables`.

`~/.ssh/config` is generated by home-manager (`hosts/gusto/ssh.nix`):

- **Global lines first**, before any `Host` block (`extraOptionOverrides`): `IdentityAgent`, `IdentitiesOnly`, `Compression`, `SendEnv`. The first value ssh finds wins, so no later block – and no included file – can point the agent elsewhere. Exactly that had broken the old config: an included file's `Host *` sent IdentityAgent to another user's agent.
- **One key per target**: `IdentityFile` names the *public* half (`hosts/gusto/ssh/*.pub`, linked to `~/.ssh/pub/`). That tells the agent which key to offer; the private half never touches the disk.
- **Lab hosts are generated** from `inventory.json` – every managed host except gusto: `<host>` / `<host>.peer.homelab.example` over the mesh as root (zorlock as admin); `<host>.example-tailnet.ts.net` while Tailscale still runs; and for hosts with a public address (aldrin, grammi, zummi) `<host>.homelab.example` as admin – root is refused on public addresses (`modules/ssh.nix`), so this fallback works when the mesh is down. The alternative names use `HostKeyAlias <host>.peer.homelab.example`, so all of them are checked against the one host key from the inventory.
- **Host keys** come from the inventory (`/etc/ssh/ssh_known_hosts`, like on every host); GitHub's key is pinned in `configuration.nix`. `just known-hosts` is not needed on gusto.
- By hand: `zorlock-lan` (FileVault unlock), `opnsense`, `arkkvm`/`arkkvm01`, the Subraum router as jump host (with its port forwards), vogon, amnesia, unifi, git.c3pb.de, edi, a block for old switches that only *adds* legacy algorithms (`+…`), and GitHub (with a 60 s ControlMaster).

**devShell and direnv**

The repo's own tools are not installed globally. `flake.nix` has `devShells.aarch64-darwin.default` with sops, age, ssh-to-age, opentofu and nvd from the flake's pins; `.envrc` (`use flake`) makes direnv load it whenever a shell enters the checkout – once per checkout `direnv allow`. nix-direnv caches the shell and protects it from garbage collection. The docs tools are not in it: `just docs` and `just docs-serve` fetch their own through `nix shell`.

sops on gusto takes the workstation's age key straight from 1Password: home-manager sets `SOPS_AGE_KEY_CMD='op read "op://<vault>/<item>/password"'`, `op` unlocks with Touch ID through the app – no `op signin`, no key file.

**DNSControl**

The public DNS records live in the repo next door, `~/Projects/homelab/DNSControl` (local, no remote; linked in here as the `dnsconfig.js` symlink). dnscontrol comes from a second, slim devShell `devShells.aarch64-darwin.dns` that holds nothing but dnscontrol and sops; the `.envrc` there loads it with `use flake ../nixlab#dns`. DNSControl gets no `flake.nix` of its own – that keeps dnscontrol on this flake's pins, with no second `flake.lock`.

The providers' credentials (deSEC, OVH, hosting.de) live sops-encrypted in `creds.sops.json` in the repo, never in plaintext on disk. The `.sops.yaml` there names the same recipients as the Tofu modules: the workstation key and the `admin` recovery key – so they stay readable without 1Password. A wrapper does the rest:

- `bin/dnscontrol` sits in front of the real binary through the `.envrc` (`PATH_add bin`), which stores the real one's path in `DNSCONTROL_BIN`.
- For the commands that talk to providers – `preview`, `push`, `ppreview`, `ppush`, `check-creds`, `get-zones`, `get-certs` – it adds `--creds '!sops -d …/creds.sops.json --output-type json'`, unless `--creds` is on the line already. Everything else it passes through unchanged. `--output-type` comes last on purpose: dnscontrol takes a `--creds` value ending in `.json` for a file name, even after the `!`.
- sops decrypts with the workstation key (`SOPS_AGE_KEY_CMD`, Touch ID), dnscontrol reads the JSON from its output. None of it touches the disk.

The commands stay `dnscontrol preview` and `dnscontrol push`. Changing a token means `sops creds.sops.json` and a commit. The repo's `.gitignore` blocks a `creds.json`, so plaintext never gets in again. An executable `creds.json` instead of the wrapper does not work: dnscontrol does not check the executable bit for names ending in `.json`.

## Add a new host

A host appears in several places in the repo, because NixOS and OpenTofu deliberately share no host list (see What is deliberately not wired). This guide walks through them in order.

### 1. Provide the machine

Rented VMs are created by hand at the provider; bare metal and the Subraum VM follow the installation guides in `docs/install-*.md`. The result is a NixOS install with an SSH host key – the host's age key is derived from it next.

### 2. Create the configuration

- Create `hosts/<host>/configuration.nix` and `hardware-configuration.nix`.
- Add it to `nixosConfigurations` in `flake.nix` with an explicit module list. Whatever is left out on purpose gets a comment saying why.

### 3. Add it wherever hosts are listed

| File | Entry |
|---|---|
| `flake.nix` | `nixosConfigurations.<host>` |
| `inventory.json` | `hosts.<host>` with `"managed": true` – kind, site, owner, unlock (see Inventory) |
| `justfile` | `hosts := "…"` (pull hosts only, for `ship-all`) |
| `terraform/netbird/main.tf` | `local.hosts`: host → setup key group (`servers` or similar) |
| `terraform/desec-tokens/main.tf` | `local.hosts`: allowed source addresses, `acme_names` |
| `terraform/github/main.tf` | `local.hosts`: deploy key read-only or with write access |

### 4. sops recipient

Derive the age key from the new host's SSH host key (`ssh-to-age` on `/etc/ssh/ssh_host_ed25519_key.pub`) and add it to `.sops.yaml`: an anchor under `keys` and a `creation_rule` for `hosts/<host>/(secrets|netbird|desec|github)\.yaml$` with the host and `admin` as recipients. The public host key itself goes into `inventory.json` (`sshHostKeys`, or later via `just inventory-collect`); `checks.inventory-sops` compares the two.

### 5. Mint keys and tokens

| Recipe | Result | Note |
|---|---|---|
| `just netbird-key <host>` | `hosts/<host>/netbird.yaml` | one-off setup key, **valid for 24 h** – deploy right after |
| `just desec-key <host>` | `hosts/<host>/desec.yaml` | only for hosts with their own certificates |

| Recipe | Result | Note |
|---|---|---|
| `just github-key <host>` | `hosts/<host>/github.yaml` + `github-deploy.pub` | pull hosts only |

Recipes that need sops take the workstation key from 1Password (on gusto set up by home-manager, see just recipes).

### 6. Deploy and check

- First deploy within 24 h of `netbird-key`. The host registers itself in the mesh at startup; Tofu finds it by name through `data "netbird_peer"`.
- Run `just topology` and commit the result – otherwise `nix flake check` fails.
- `just inventory-collect` (host key, interfaces), set `network/ipv4` by hand, then `just inventory-test`; `just mac-switch` on gusto (its known_hosts comes from the inventory) – a new managed host also gets its SSH block there automatically, with `hosts/gusto/ssh/<host>.homelab.example.pub`.
- Add app secrets once the host is running: Add secrets.
- Update the docs: Hosts & roles, Services.

## Publish a new service

Services are named by function (`rezepte`, not `mealie`); new web services on grammi use PostgreSQL. First decide how the service should be reachable – that decides which files change.

| Reachability | For | Examples |
|---|---|---|
| Mesh only | admin UIs, services for your own devices | `s3-admin`, `llm` |
| Public behind the reverse proxy with SSO | web apps that should work on the road without the NetBird app | `rezepte`, `grist`, `metrics`, `dokumente` |
| Directly public | services with their own login, for machines or everyone | `auth`, `s3.aldrin` |

### Mesh only

1. One line under `local.mesh_services` in `terraform/netbird/mesh-services.tf`: name → host(s). This creates a NetBird DNS zone of its own with A record(s) on the mesh address(es); public DNS does not know the name.
2. `just netbird-apply` – since NetBird 0.81 new zones reach the clients without a restart.
3. On the host an nginx vhost with `enableACME`, `acmeRoot = null` (DNS-01 via deSEC) and `allow 192.0.2.0/24; deny all;`.
4. If the service runs on ursalia or zorlock: add the name to the host's `acme_names` in `terraform/desec-tokens/main.tf` and run `just desec-apply` – their tokens may only write exactly those TXT records.
5. Deploy the host, `just topology` (new vhost).

### Public behind the reverse proxy

For services on grammi:

1. An entry in `hosts/grammi/proxy-services.json`: name → app, port, Pocket ID groups. Tofu (`reverse-proxy.tf`) and `topology.nix` read the same file.
2. The service listens on all addresses; only 80/443 are open publicly, `nb-mesh` is trusted.
3. Push a CNAME `<name>` → `proxy.homelab.example` in DNSControl – a few minutes before the apply, otherwise the resolver caches NXDOMAIN.
4. Deploy grammi first, then `just netbird-apply`. At the end it runs `just netbird-proxy-certs`: it requests every new name once from outside so the proxy fetches its certificate, then restarts `netbird-proxy` once – otherwise the dashboard hangs at "Issuing certificate…" (NetBird discussion #7118).

Services on other hosts get a resource of their own in `terraform/netbird/reverse-proxy.tf` with the host as peer target (examples: `dokumente` on ursalia – SSO for the app, a PIN for share links only –, `doku` on aldrin). Here too: push the CNAME first, `just netbird-apply` takes care of the certificates.

### Directly public

nginx vhost on the host (ACME via DNS-01), A/AAAA records in DNSControl, CrowdSec reads the nginx logs. Only for services that have a login of their own anyway.

### Afterwards

- Docs: Services, and Contents of hosts/<host>/ if new files appear there.
- Dual stack wherever possible (A and AAAA).

## Add secrets

App passwords, OIDC clients and the like live in `hosts/<host>/secrets.yaml`, encrypted for the host and the recovery key `admin`. The MacBook cannot decrypt these files – that is why they are edited on the host (background: Secrets, network & access).

### Steps

1. `just sync <host>` – otherwise the host has an outdated `secrets.yaml`, and `pull-secrets` brings it back into the repo.
2. `just secrets <host>` – opens sops on the host (with its host key, no 1Password sign-in) and afterwards fetches the result into the repo via `pull-secrets`.
3. Always quote values: `key: "value"`, single quotes if the value contains `"` or `\`. An unquoted value may be read by YAML as a number or bool; sops-nix then aborts `setupSecrets` and **no** secret of that generation gets installed.
4. Add the NixOS side (`sops.secrets."<name>"`) in the host module, commit, deploy: `just ship <host> "<message>"`.

### What is not edited by hand

| File | Written by |
|---|---|
| `hosts/<host>/netbird.yaml` | `just netbird-key <host>` |
| `hosts/<host>/desec.yaml` | `just desec-key <host>` |
| `hosts/<host>/github.yaml` | `just github-key <host>` |

| File | Written by |
|---|---|
| `terraform/*/secrets.yaml` | sops on the MacBook with the workstation key from 1Password |

Everything that needs sops on the MacBook runs with the sign-in in front – see just recipes.

## Rotate keys

What can be renewed how, and what to do afterwards. The recipes replace the old key at once – so commit and deploy right after.

| Key | Recipe | Afterwards | When |
|---|---|---|---|
| A host's NetBird setup key | `just netbird-key <host>` | deploy within 24 h | after a reinstall or lost client state |
| A host's deSEC token | `just desec-key <host>` | deploy at once, then `just desec-verify` | on suspicion, otherwise never |
| A host's GitHub deploy key | `just github-key <host>` | commit + deploy | on suspicion |
| Tofu token for NetBird (service user `terraform`) | `just netbird-renew-token` | – | yearly |
| App secrets | `just secrets <host>` (see Add secrets) | deploy | on suspicion |

### SSH host key (= sops recipient)

A host's age key is derived from its SSH host key. A new host key therefore also means:

1. Derive the new age key with `ssh-to-age` and replace it in `.sops.yaml`.
2. Re-encrypt the host's files with `sops updatekeys` – this needs a key that can still read them (the old host key or `admin`).
3. Put the new public key into `inventory.json`: `just inventory-collect --force <host>`. `nix flake check` (`checks.inventory-sops`) fails until the inventory and `.sops.yaml` agree.
4. Commit and deploy before the old key is gone – every host's known_hosts comes from the inventory. On gusto: `just mac-switch`.

### After a rotation

- Delete old keys at the other end (GitHub, deSEC and NetBird are handled by the recipe through Tofu).
- `just docs` if names or procedures changed.

## Repository layout

One flake, `nixosConfigurations` for five hosts plus `darwinConfigurations` for the two Macs (zorlock, gusto) and two `devShells`: one for working on the repo, one with dnscontrol for the DNSControl repo. Every host gets an explicit module list in `flake.nix`; whatever is missing is missing with the reason in a comment next to it.

| Path | Contents |
|---|---|
| `flake.nix` / `flake.lock` | Inputs, host definitions, packages, `lib.topology`, `lib.inventory`, `checks` |

| Path | Contents |
|---|---|
| `modules/` | 17 shared modules: base, nix, ssh, firewall, networking (ifstate), acme, crowdsec, tailscale, netbird, metrics/metrics-shipping, logging/log-shipping, qemu-guest, deploy, auto-upgrade-git, docs-site |
| `modules/darwin/` | For the Macs: `homebrew.nix` (one Homebrew policy for every darwin host) and `rebuild.sh` (the engine behind the `mac-*` recipes) |
| `hosts/gusto/` | The workstation: `configuration.nix` (system, casks, macOS settings), `home.nix` and `ssh.nix` (home-manager), `ssh/*.pub` (public halves of the 1Password SSH keys) |
| `.envrc` | direnv: loads the flake's devShell (sops, age, tofu, …) in the checkout |
| `hosts/<host>/` | `configuration.nix`, hardware, service modules, sops files (`secrets.yaml`, `netbird.yaml`, `desec.yaml`) |
| `pkgs/` | Own packages: garage-ui, logchef, paperless-gpt, prometheus-paperless-exporter, ollama-metrics-proxy |
| `terraform/netbird/` | OpenTofu: NetBird account, policies, DNS zones, reverse proxy services, people/groups (incl. Pocket ID) |
| `terraform/desec-tokens/` | OpenTofu: narrowly scoped deSEC tokens per host for DNS-01 |
| `terraform/github/` | OpenTofu: the repositories `nixlab` (private) and `nixlab-docs` (public), deploy keys per host |
| `topology.nix` | Generates the Grafana topology dashboard from the host configs |
| `inventory.json` / `inventory.nix` | Lab inventory (every machine: kind, site, uplink, relations, unlock) and its validation, see Inventory |
| `inventory.schema.json` | JSON Schema for editing `inventory.json`, generated |
| `tests/` | Tests evaluated by `nix flake check` (`inventory.nix`) |
| `justfile` | All operational workflows: deploy, secrets, netbird-*, desec-* |
| `docs/` | This project documentation: pages under `site/{de,en}/`, Zensical config, build (`build.sh`, `site-src.sh`, `assemble.py`, `gen-reference.py`), the private site's package (`package.nix`) and what it would otherwise load from third parties – fonts, glightbox (`assets.nix`), install guides (ursalia, cuckoosegg, gusto), drafts of upstream issues |
| `dnsconfig.js` | Symlink into the separate DNSControl repo – public DNS records live there |
| `.sops.yaml` | age recipients per host, `admin` (offline recovery) and `workstation` (1Password) |

## Where do I change what?

Which file is responsible for which change, how often that happens, and what to do afterwards.

| What | Where | How often | Afterwards |
|---|---|---|---|
| A service, package or option on one host | `hosts/<host>/*.nix` | often | `just ship <host> "…"` |
| Something for all hosts | `modules/*.nix` | now and then | `just ship-all "…"`; new module: public or in `docs/modules-private.txt`? |
| A name behind the reverse proxy (grammi) | `hosts/grammi/proxy-services.json` | rarely | CNAME in DNSControl, deploy grammi, `just netbird-apply` |

| What | Where | How often | Afterwards |
|---|---|---|---|
| A mesh-only name | `terraform/netbird/mesh-services.tf` | rarely | `just netbird-apply` |
| Who may read the private docs | group `admin-workstations` (`terraform/netbird/subraum.tf`, peer names) | rarely | `just netbird-apply`, commit `admin-peers.json`, deploy aldrin and ursalia |
| NetBird policies, groups, people, routes | `terraform/netbird/*.tf` | rarely | `just netbird-plan`, `just netbird-apply` |
| Public DNS records | DNSControl repo (`dnsconfig.js`) | rarely | `dnscontrol preview`, `dnscontrol push` there, commit |
| What a deSEC token may do | `terraform/desec-tokens/main.tf` | rarely | `just desec-apply` |
| Repositories, deploy keys | `terraform/github/` | rarely | `just github-plan`, `just github-apply` |
| Version of an automatically updated input | `flake.lock` | nightly, automatic | – |
| Version of a separately pinned input | `flake.lock` (`nix flake update <input>`) | by hand | deploy the hosts that use it |
| Grafana dashboard | `hosts/grammi/dashboards/*.json` | now and then | deploy grammi |
| Topology (new host, exporter, vhost) | generated by `topology.nix` | when needed | `just topology`, commit |
| Machine data: kind, hardware, site, uplink, power, unlock, networks, interfaces, host keys | `inventory.json` | with every new device | `just inventory-collect`, `just inventory-test` (Inventory) |
| An app or CLI tool on the workstation | `hosts/gusto/configuration.nix` (cask) or `hosts/gusto/home.nix` (CLI) | when needed | `just mac-diff`, `just mac-switch` |
| Homebrew policy of the Macs | `modules/darwin/homebrew.nix` | rarely | `just mac-switch` on every Mac with Homebrew |
| Versions of the casks | `flake.lock` (Homebrew inputs) | by hand | `just mac-upgrade`, commit |
| Inventory format (fields, kinds) | `spec` in `inventory.nix` | rarely | `just inventory-schema`, test case, docs |
| A host's secrets | `hosts/<host>/secrets.yaml` | rarely | Add secrets |
| This documentation | `docs/site/{de,en}/` | with every change | `just docs` |

## Contents of hosts/<host>/

What a host directory contains and who writes it. Not every host has every file.

| File | Content | Written by |
|---|---|---|
| `configuration.nix` | host configuration: network, services, vhosts | by hand |
| `hardware-configuration.nix` | hardware, file systems | `nixos-generate-config`, then by hand |
| `disk-config.nix` | partitioning (disko, ursalia only) | by hand |

| File | Content | Written by |
|---|---|---|
| further `*.nix` | one service per file, e.g. `backup.nix`, `samba.nix`, `netbird.nix`, `traefik.nix` | by hand |
| `secrets.yaml` | app secrets, encrypted for host + admin | `just secrets <host>` |
| `netbird.yaml` | NetBird setup key | `just netbird-key <host>` (Tofu) |
| `desec.yaml` | deSEC token for DNS-01 | `just desec-key <host>` (Tofu) |
| `github.yaml` | private half of the deploy key | `just github-key <host>` |
| `github-deploy.pub` | public half, read by `terraform/github` | `just github-key <host>` |
| further `*.yaml` | sops files of single services, e.g. `netbird-server.yaml`, `netbird-proxy.yaml` (grammi), `push-cuckoosegg.yaml` (ursalia) | their recipes, e.g. `just netbird-server-secrets`, `just netbird-proxy-token` |
| `push-key.pub` | public key for the push deploy (cuckoosegg only) | `just cuckoosegg-push-key` |
| `dashboards/*.json` | Grafana dashboards (grammi only) | by hand, `topology.json` via `just topology` |
| `proxy-services.json` | services behind the reverse proxy (grammi only) | by hand, read by Tofu and `topology.nix` |

## Inputs and their pins

Only `nixpkgs`, `nixpkgs-unstable` and `sops-nix` move with the nightly auto-upgrade (`updateInputs` in `modules/deploy.nix`). Anything that would lock the whole lab out on a broken release is pinned separately and bumped by hand.

| Input | Branch / tag | Used by | Nightly? |
|---|---|---|---|
| `nixpkgs` | nixos-26.05 | all NixOS hosts | yes |
| `nixpkgs-unstable` | nixos-unstable | grammi (Grist), ursalia (Paperless 3.x), zorlock (mactop) | yes |
| `sops-nix` | main | all | yes |
| `nixpkgs-netbird` | nixos-unstable | NetBird server and all clients | no – server first, then clients |
| `nixpkgs-pocketid` | nixos-26.05 | zummi (pocket-id package) | no |
| `disko` | main | ursalia (all mounts) | no |
| `lanzaboote` | v1.1.0 | ursalia (Secure Boot) | no |
| `nixpkgs-darwin`, `nix-darwin` | 26.05-darwin | zorlock, gusto | no (no auto-upgrade) |
| `home-manager` | release-26.05 | gusto (dotfiles) | no – together with `nix-darwin` |
| `nix-homebrew` | main | Macs with Homebrew (gusto): Homebrew itself | no – `just mac-upgrade` |
| `homebrew-core`, `homebrew-cask` | main (`flake = false`) | Macs with Homebrew: the pinned taps | no – `just mac-upgrade` |
| `netbirdio-tap` | main (`flake = false`) | gusto: NetBird app (cask `netbird-ui`) | no – `just mac-upgrade` |

The Homebrew inputs are the reason a cask has a fixed version: with `mutableTaps = false` Homebrew cannot update its taps itself, see macOS hosts.

`nix flake check` additionally verifies that `hosts/grammi/dashboards/topology.json` matches what `topology.nix` currently generates – a new exporter without `just topology` fails the check.

## Hosts & roles

Three rented VMs carry the public services, ursalia and zorlock sit at home, cuckoosegg in the Subraum hackerspace; gusto is the admin workstation (a MacBook). All servers are reachable in the NetBird mesh as `<host>.peer.homelab.example`; root SSH exists only there. Kind, site, uplink and disk unlock of every machine – including the routers, KVM and appliances – are in the inventory.

| Host | Platform | Role | Notes |
|---|---|---|---|
| grammi | NixOS, rented VM | NetBird server (`mesh.homelab.example`) + reverse proxy, Traefik, monitoring (VictoriaMetrics/-Logs, Grafana, logchef), Mealie, Grist, PostgreSQL | Central sink for metrics and logs of all hosts |
| zummi | NixOS, rented VM | Pocket ID (`auth.homelab.example`) – login for all services and NetBird | pocket-id pinned separately |
| aldrin | NixOS, rented VM | Garage S3 (`s3.aldrin.homelab.example`), garage-ui (`s3-admin`), backup target for restic, docs site (`doku`) | Forked garage-ui package with subpath support |
| ursalia | NixOS, bare metal (ThinkCentre M720q) | Paperless-ngx 3 + paperless-gpt (`dokumente`), docs site (`doku`), Samba for the scanner, routing peer into the home LAN, build host for cuckoosegg, Claude Code bridge to the NanoKVM Go+ (on demand) | LUKS + btrfs via disko, TPM unlock, Secure Boot (lanzaboote), WLAN (Intel BE200), no CrowdSec |
| cuckoosegg | NixOS, VM on the Subraum Proxmox | NetBird routing peer into the Subraum networks | Group `subraum-devices` instead of `servers`, no repo/deploy key, deployed by push |
| zorlock | macOS + nix-darwin, Mac mini M4 | Ollama on the Apple GPU (`llm`), used by paperless-gpt | None of the `modules/*.nix`; FileVault unlock after reboot via LAN SSH |
| gusto | macOS + nix-darwin + home-manager, MacBook Pro | Admin workstation: deploys, secrets, Tofu, docs | Laptop, no sshd, no secrets, not monitored; apps as Homebrew casks, NetBird as the official app – see gusto and macOS hosts |

## gusto – the admin workstation

gusto is the MacBook Pro everything in this repo is run from: deploys, secrets, Tofu, the docs. It is the second nix-darwin host next to zorlock, and the only machine in the lab a person sits at. This page is the complete reference for it – what is on it, where each part is configured, how it is used day to day and what to check when something does not work. *Why* it is built this way (Homebrew policy, NetBird as an app, the differences to zorlock) is on macOS hosts.

### At a glance

| | |
|---|---|
| Hardware | Apple MacBook Pro (Apple silicon), `kind: laptop`, `availability: intermittent` in the inventory |
| System | macOS (English, region Germany), FileVault on |

| | |
|---|---|
| Configuration | `darwinConfigurations.gusto` in `flake.nix`: nix-darwin + home-manager (user `admin`) |
| Built and switched | on gusto itself, from the working copy `~/Projects/homelab/nixlab` – `just mac-switch` |
| Apps | Homebrew casks, Homebrew and its taps pinned in `flake.lock` (`modules/darwin/homebrew.nix`) |
| Repo tools | from the flake's devShell, loaded by direnv in the checkout |
| SSH | client only, keys only in 1Password; no Remote Login (no sshd) |
| Mesh | NetBird peer `gusto` (cask `netbird-ui`), group `admin-workstations` |
| Secrets | none of its own – sops on gusto *uses* the workstation age key from 1Password |
| Monitoring | none; not in the topology, nothing in the lab depends on it (`criticality: lab`) |
| Backup | Arq 7 to Garage on aldrin, plus Time Machine |

## Files

| File | Content |
|---|---|
| `hosts/gusto/configuration.nix` | System: host name, Nix settings, Touch ID for sudo, casks, system packages, font, SSH known hosts, `SSH_AUTH_SOCK` for GUI apps, macOS settings, home-manager wiring |
| `hosts/gusto/home.nix` | admin's home: environment, links, zsh and its helpers, CLI/TUI tools, Ghostty, git |
| `hosts/gusto/ssh.nix` | `~/.ssh/config` and `~/.ssh/pub/` |
| `hosts/gusto/ssh/*.pub` | Public halves of the SSH keys (one per target), linked to `~/.ssh/pub/` |
| `modules/darwin/homebrew.nix` | Homebrew policy shared with zorlock |
| `modules/darwin/rebuild.sh` | The engine of the `mac-*` recipes |
| `.envrc` | `use flake` – direnv loads the devShell in the checkout |
| `docs/install-gusto.md` | Setting gusto up from scratch (private docs only) |

Flake inputs used only by gusto: `home-manager`, `nix-homebrew`, `homebrew-core`, `homebrew-cask`, `netbirdio-tap` – see Flake inputs.

## What is installed

gusto was set up deliberately bare on 2026-10-21: no app list taken over from the previous installation, only the base below. Anything else is added when it is needed (see Day to day).

### Apps (Homebrew casks)

| Cask | What for |
|---|---|
| `1password`, `1password-cli` | Passwords, the SSH agent with all SSH keys, `op` for scripts (sops' age key). The CLI as a cask, not from nixpkgs: it unlocks through the app with Touch ID, which only works with the vendor's signed binary. |
| `vivaldi` | Browser |
| `arq` | Backup to Garage on aldrin |
| `claude` | Claude desktop app, incl. the local MCP servers (Feedbin, arkkvm, MoneyMoney, Home Assistant) |
| `ghostty` | Terminal |

| Cask | What for |
|---|---|
| `netbirdio/homebrew-tap/netbird-ui` (+ formula `netbird`) | NetBird menu-bar app and daemon – see NetBird |
| `shottr` | Screenshots |
| `telegram`, `whatsapp` | Messengers |
| `spotify` | Music |
| `visual-studio-code` | Editor (Microsoft's build, for Remote-SSH and Pylance); telemetry off, `settings.json` comes from home-manager and is read-only |

From the Mac App Store (`homebrew.masApps`, through `mas`; needs the App Store sign-in): **TestFlight**. `zap` does not remove App Store apps – a deleted line leaves the app installed.

### System-wide (nix-darwin)

`git`, `gh`, `just`, `jq`, `neovim` (`EDITOR`/`VISUAL`, kept across sudo), `curl`, `wget`, `rsync`; zsh as the system shell (`programs.zsh`); the font **Maple Mono NF** (Nerd Font) for Ghostty and the TUIs.

### admin (home-manager)

| Tool | Notes |
|---|---|
| zsh | config in `~/.config/zsh`, autosuggestions, syntax highlighting, history in `~/.local/state/zsh/history` |
| atuin | shell history in SQLite, Ctrl-R / ↑ search; update check off |
| fzf | Ctrl-T (files), Alt-C (cd) |
| zoxide | `z <part of a path>` |
| direnv + nix-direnv | per-directory environments; caches flake shells and protects them from GC |
| bat, eza, ripgrep, fd, tree | `cat`/`ls`/`grep`/`find` replacements; eza brings `ls`, `ll`, `la`, `lt` |
| lazygit, btop, yazi, tmux | TUIs; yazi as `y` (changes into the last directory on exit), tmux with mouse, `tmux-256color` |
| nvd, nix-output-monitor | closure diffs, readable build output (`nom`) |
| uv | Python tools (`uv tool install` → `~/.local/bin`) |
| typst | documents (real-fast-document) |

### Repo tools (devShell)

`sops`, `age`, `ssh-to-age`, `opentofu`, `nvd` – from `devShells.aarch64-darwin.default`, at the versions in `flake.lock`. They are there in every shell inside the checkout (direnv), not elsewhere. `dnscontrol` comes from `devShells.aarch64-darwin.dns` and is only there in the DNSControl folder, behind a wrapper that decrypts the credentials with sops – see macOS hosts. The docs tools (pandoc, WeasyPrint, Zensical) are not in it; `just docs` and `just docs-serve` fetch their own.

## Environment

| Variable / path | Value | Why |
|---|---|---|
| `PATH` | Nix profiles, then `/opt/homebrew/bin:sbin`, then macOS' directories; `~/.local/bin` in front for admin | Nix wins over Homebrew, Homebrew over Apple's stubs; no `brew shellenv` anywhere |
| `SSH_AUTH_SOCK` | `~/.1password/agent.sock` | every SSH client talks to 1Password – in shells (home-manager) and GUI apps (`launchd.user.envVariables`) |
| `SOPS_AGE_KEY_CMD` | `op read "op://<vault>/<item>/password"` | sops gets the workstation key from 1Password when it needs it; never on disk |
| `EDITOR`, `VISUAL` | `nvim` | |
| `GOPATH` / `GOBIN` | `~/.local/share/go` / `~/.local/bin` | XDG; binaries into the one personal PATH directory |
| `CARGO_HOME` / `CARGO_INSTALL_ROOT` | `~/.local/share/cargo` / `~/.local` | same; `cargo install` writes to `<root>/bin` |
| `ZDOTDIR` | `~/.config/zsh` | set by the stub `~/.zshenv` home-manager writes |

### Files and links home-manager manages

| Path | |
|---|---|
| `~/.zshenv` → `~/.config/zsh/.zshenv`, `.zshrc` | zsh |
| `~/.config/git/config` | git |
| `~/.config/ghostty/config` | Ghostty |
| `~/.config/tmux/tmux.conf`, `~/.config/atuin/config.toml`, `~/.config/direnv/…`, `~/.config/lazygit/config.yml` | the tools |
| `~/.ssh/config`, `~/.ssh/pub/` | SSH |
| `~/.1password/agent.sock` → `~/Library/Group Containers/2BUA8C4S2C.com.1password/t/agent.sock` | path without spaces for the agent; 1Password does not create it |
| `~/.config/ssh` → `~/.ssh` | XDG rule: tools without XDG support get a link under `~/.config` |

A file home-manager wants to write but finds already there (first switch on a fresh macOS) is renamed to `<name>.before-hm` instead of failing the switch.

## macOS settings

Decided one by one on 2026-10-21; everything not listed is macOS' default.

| Area | Setting | nix-darwin option |
|---|---|---|
| Menu bar clock | 24 h, weekday, no date | `menuExtraClock.Show24Hour`, `ShowDayOfWeek`, `ShowDate = 2` |
| Dock | always visible, no magnification, icon size 66 | `dock.autohide`, `magnification`, `tilesize` |
| | no launch animation, no bouncing | `dock.launchanim`, `CustomUserPreferences."com.apple.dock"."no-bouncing"` |

| Area | Setting | nix-darwin option |
|---|---|---|
| | minimise into the app icon, effect "scale", no recent apps | `minimize-to-application`, `mineffect`, `show-recents` |
| | hot corner bottom right: Quick Note | `dock.wvous-br-corner = 14` |
| | Dock content | not managed (yet) |
| Finder | path bar on, status bar off, icons on the desktop, list view, folders first (windows and desktop) | `finder.ShowPathbar`, `ShowStatusBar`, `CreateDesktop`, `FXPreferredViewStyle = "Nlsv"`, `_FXSortFoldersFirst`, `_FXSortFoldersFirstOnDesktop` |
| General | dark appearance, all file extensions, auto-capitalisation, period on double space | `NSGlobalDomain.AppleInterfaceStyle`, `AppleShowAllExtensions`, `NSAutomaticCapitalizationEnabled`, `NSAutomaticPeriodSubstitutionEnabled` |
| Trackpad | Force Click on, tracking speed 1 | `NSGlobalDomain."com.apple.trackpad.forceClick"`, `"com.apple.trackpad.scaling"` |
| Keyboard | German (main) and U.S. input sources | `CustomUserPreferences."com.apple.HIToolbox".AppleEnabledInputSources` – U.S. for KVM sessions (NanoKVM Go+, ArkKVM type PC scancodes); effective after logging out once |
| Network names | `gusto` as computer, local host and SMB NetBIOS name | `networking.*Name`, `smb.NetBIOSName` |
| Language, region | English, Germany | not in Nix – set during macOS setup |

After each switch `activateSettings -u` runs as admin, so most settings apply without logging out.

## Nix

Experimental features `nix-command flakes`; `admin` is a trusted user; no channels; `min-free` 5 GiB / `max-free` 20 GiB; store optimisation on; garbage collection Sundays 12:15, everything older than 30 days. No extra binary caches – every input follows nixpkgs, a community cache would have no hits.

## Security

- **FileVault** on (macOS setup).
- **sudo with Touch ID** (`/etc/pam.d/sudo_local`, which macOS updates leave alone), working inside tmux too (`pam_reattach`).
- **No Remote Login**: gusto runs no sshd and accepts no SSH connections. Nothing in the inventory or in `.sops.yaml` has a key for it, so no secret is ever encrypted to it.
- **Keys and secrets only in 1Password**: SSH keys (agent), the sops age key (`op read`), the Arq credentials. Nothing private under `~/.ssh`.

## SSH

`~/.ssh/config` is generated (`hosts/gusto/ssh.nix`). At the very top, before any `Host` block: `IdentityAgent ~/.1password/agent.sock`, `IdentitiesOnly yes`, `Compression yes`, `SendEnv LANG LC_*` – first value wins in ssh_config, so nothing below can change them. Every `IdentityFile` names a **public** key in `~/.ssh/pub/`; that is how the agent knows which single key to offer.

Host keys: lab hosts from `inventory.json` (`/etc/ssh/ssh_known_hosts`), GitHub pinned; the rest goes to `~/.ssh/known_hosts` on first use.

| ssh … | Goes to | User | Key (`~/.ssh/pub/`) | Notes |
|---|---|---|---|---|
| `aldrin`, `grammi`, `zummi`, `ursalia`, `cuckoosegg` (also `<host>.peer.homelab.example`) | `<host>.peer.homelab.example` (NetBird) | root | `<host>.homelab.example.pub` | generate inventory managed gusto |
| `zorlock` | `zorlock.peer.homelab.example` | admin | `zorlock.homelab.example.pub` | no root o |
| `<host>.example-tailnet.ts.net` | Tailscale | as above | as above | break-gla Tailscale not for cu Goes awa Tailscale. |
| `aldrin.homelab.example`, `grammi.homelab.example`, `zummi.homelab.example` | public address | admin | as above | fallback mesh is is refuse addresse `ssh.nix` |
| `zorlock-lan` | 198.51.100.86 | admin | `zorlock.homelab.example.pub` | FileVault a reboot, home LA |
| `opnsense` | 198.51.100.1 | root | `opnsense.pub` | |
| `arkkvm`, `arkkvm01` | `arkkvm01` (DHCP name) | root | `arkkvm.pub` | |
| `subraum-router` | hackspace.c3pb.de | admin | `c3pb.de.pub` | jump hos 8006 → P 8080/808 Assistant |
| `vogon`, `amnesia`, `unifi` | `<name>.subraum.c3pb.de` via `subraum-router` | root | `<name>.subraum.c3pb.de.pub` | |
| `git.c3pb.de` | | git | `git.c3pb.de.pub` | |
| `edi` | | admin | – | `HostKey` `+ssh-rsa` |
| `switch-*`, `172.31.64.*`, `172.31.65.*` | | bernd (switches) | – | old switc SHA-1 KE CBC ciph |
| `github.com` | | git | `github.com.pub` | ControlM |
| `github-443` | ssh.github.com:443 | git | `github.com.pub` | for netwo block por |

The alternative names (tailnet, public) carry `HostKeyAlias <host>.peer.homelab.example`, so all of them are checked against the one host key in the inventory. A new managed host appears here by itself after `just mac-switch` – only its public key file has to be added to `hosts/gusto/ssh/`.

### git

`~/.config/git/config`: name `admin`, address `mail@admin.org`; below `~/Projects/homelab/` (`includeIf gitdir:`) `admin@homelab.example`. `init.defaultBranch = main`, `pull.rebase = true`, alias `lg = log --oneline`. GitHub over SSH with the key `github.com SSH (admin)` from 1Password; `gh` for the API.

### NetBird

The official app from NetBird's tap (cask `netbird-ui`, formula `netbird` for the daemon), pinned as the input `netbirdio-tap`. Sign-in by SSO (Pocket ID) in the app, peer name `gusto`, group `admin-workstations` (Subraum devices, docs site). Rosenpass is **off** on the workstation for now (test since 2026-10-21). The version follows the tap, not `nixpkgs-netbird`; `just mac-upgrade` prints both. Details and the comparison with zorlock: macOS hosts.

### Backup

- **Arq 7** backs up the whole disk to the Garage bucket on aldrin, with its own backup plan for gusto (the previous MacBook's plan is kept for restores only). Encryption password and Garage credentials are in 1Password.
- **Time Machine** in addition.
- Nothing on gusto needs a backup to be rebuilt: the system and the dotfiles are this repo, the keys are in 1Password. A backup is for data.

### Day to day

| Task | How |
|---|---|
| Add an app | cask in `hosts/gusto/configuration.nix` → `just mac-diff` → `just mac-switch` |
| Add a CLI tool | `programs.<tool>` or `home.packages` in `hosts/gusto/home.nix` → `just mac-switch` |
| Remove something | delete the line → `just mac-switch`; casks go with their data (`zap`) – `just mac-diff` shows it first |
| Update apps (Homebrew) | `just mac-upgrade` – bumps only the tap inputs, switches, `brew upgrade`, summary; then commit `flake.lock` |
| Update the system (Nix) | `nix flake update nixpkgs-darwin nix-darwin home-manager` → `just mac-switch` → also `just mac-switch zorlock` (same inputs) |
| Deploy a NixOS host | `just ship <host> "<message>"` or `just deploy <host>` – over the mesh as root |
| Edit secrets | `just secrets <host>` (Touch ID for `op`) |
| Docs | `just docs-serve` (preview), `just docs` (PDFs + public repo) |
| New checkout or worktree | `direnv allow` once inside it |

All `mac-*` recipes take the host as an optional parameter (default `gusto`; `zorlock` runs over SSH after a `git pull` there). New files must be known to git (`git add`) before a switch sees them.

### Troubleshooting

| Symptom | Check |
|---|---|
| `ssh-add -L`: "The agent has no identities" / "Could not open a connection" | 1Password running and unlocked, Settings → Developer → *Use the SSH agent*; `ls -l ~/.1password/agent.sock`; `echo $SSH_AUTH_SOCK` |

| Symptom | Check |
| --- | --- |
| ssh offers the wrong key / "Too many authentication failures" | `ssh -G <host> \| grep -E 'identityfile\|identityagent'` – exactly one `.pub` from `~/.ssh/pub/` |
| Host key mismatch for a lab host | host reinstalled or key rotated: inventory updated? Then `just mac-switch` (known hosts are part of the system) |
| `sops`, `tofu` not found | not in the checkout, or `direnv allow` missing |
| sops: "failed to get the data key" | `op read "op://<vault>/<item>/password" \| age-keygen -y` must print the workstation's `age1…` |
| `brew` installs nothing / tap missing | taps come only from `flake.lock` (`mutableTaps = false`); `brew tap` and `brew update` are not the way – add an input, `just mac-switch` |
| A setting did not apply | log out and in once (input sources, some Dock/Finder keys) |
| `.before-hm` files in the home directory | left by the first switch; compare and delete |
| NetBird disconnected | menu-bar app → Connect; `netbird status -d`; daemon: `sudo launchctl print system/netbird` |

# Inventory

`inventory.json` lists every machine of the lab, whether this flake configures it or not: what kind of device it is (vendor, model, serial), where it stands, how that place reaches the internet, what it runs on or is built into, where its power comes from, how its disk gets unlocked after a reboot, its networks, interfaces and addresses, and its SSH host keys. These are the facts a NixOS config does not hold. Today the host keys are used: every host's known_hosts comes from here, and `nix flake check` ties them to the age keys in `.sops.yaml`. Later the rest is meant for grouping in the topology and for alert dependencies (an uplink is down, so don't page for every host behind it).

| File | Role |
| --- | --- |
| `inventory.json` | The data. JSON, so Nix (`builtins.fromJSON`) and OpenTofu (`jsondecode`) can both read it without help. |
| `inventory.nix` | Format (`spec`), validation, derived view, JSON Schema generator. Exposed as `lib.inventory` in `flake.nix`. |
| `inventory.schema.json` | JSON Schema for the editor (completion, hover help, inline errors). Generated, never edited. |
| `tests/inventory.nix` | Tests for the validator: a complete example inventory and one case per rule. |
| `flake.nix` | `lib.inventory`, the checks, and `programs.ssh.knownHosts` for every host (NixOS and nix-darwin). |

Rule of thumb for what goes in: **only what cannot be derived from a config.** OS, architecture, CPU/RAM, mesh address, exporters and backup targets stay out; `topology.nix` reads those from `nixosConfigurations`. Every relation is stored in **one direction only**; the reverse is derived, so the two cannot disagree.

## Structure

Five sections, each an object of entries keyed by id. Ids are lower-case letters, digits and dashes (usable unchanged as Graphviz node or label values).

```
owners    ◄──────── hosts.owner
uplinks   ◄──────── sites.uplinks[]
sites     ◄──────── hosts.site, networks.site
networks  ◄──────── hosts.interfaces.<name>.network
hosts     ◄──────── hosts.runsOn / inside / power / manages[] / unlock.tang[], networks.dhcpServer
```

A `"$schema": "./inventory.schema.json"` key at the top is allowed and ignored; it tells the editor where the schema is.

### owners

Who administers a device: it decides whom to call and what may be changed.

| Field | Type | | Meaning |
|---|---|---|---|
| `name` | string | required | Display name |
| `contact` | string | | Who to talk to when this owner's devices are affected |
| `note` | string | | Free text |

### uplinks

An internet line. Several sites may share one (rarely), one site may have several (primary and backup).

| Field | Type | | Meaning |
|---|---|---|---|
| `isp` | string | | Provider; for a data centre the hoster |
| `asn` | integer ≥ 1 | | AS number of the provider's network, without "AS" |
| `medium` | `fiber` `dsl` `cable` `lte` `satellite` `datacenter` | | Kind of line |
| `bandwidth` | `{ "down": int, "up": int }` | | Booked bandwidth in Mbit/s, both values required |
| `ipv4` | `public` `cgnat` `none` | | What the line gives for IPv4 |
| `ipv4Static` | boolean | | Fixed public IPv4 address (only meaningful with `public`) |
| `ipv6` | boolean | | Native IPv6 |
| `ipv6PrefixLength` | integer 1–128 | | Size of the delegated IPv6 prefix (56 for a /56); only with `ipv6` |
| `ipv6Static` | boolean | | Fixed IPv6 prefix; only with `ipv6` |
| `note` | string | | Free text |

All fields are optional, so a line can be entered before its details are known.

### sites

A place where devices stand.

| Field | Type | | Meaning |
|---|---|---|---|
| `city` | string | required | City |
| `kind` | `residential` `hackerspace` `datacenter` | required | Kind of place |

| Field | Type | | Meaning |
|---|---|---|---|
| `provider` | string | | Who runs the place (hoster, club); empty for a home |
| `uplinks` | list of uplink ids | required, ≥ 1 | Internet lines, **primary first**, backups after it |
| `note` | string | | Free text |

### hosts

Every device with an address that matters: servers, VMs, routers, APs, KVMs, PDUs, appliances.

| Field | Type | | Meaning |
|---|---|---|---|
| `kind` | see kinds | required | What the device is |
| `vendor` | string | | Manufacturer (Lenovo, Apple, GL.iNet). Not for `vm`/`container` |
| `model` | string | | Model without the vendor (ThinkCentre M720q), or for a VM the provider's product/plan name |
| `productId` | string | | The vendor's id for drivers and firmware: Lenovo machine type (`10T7…`), Apple model identifier (`Mac16,10`). Not for `vm`/`container` |
| `serial` | string | | Serial number, unique lab-wide. Not for `vm`/`container` |
| `site` | site id | required* | Where it is. *Optional with `runsOn` or `inside`: then it is inherited |
| `owner` | owner id | required | Who administers it |
| `managed` | boolean | required | Configured by this flake – must match `flake.nix` |
| `criticality` | `core` `service` `lab` | default `service` | `core`: the lab stops working without it; `service`: a service is down; `lab`: nothing depends on it |
| `availability` | `always-on` `intermittent` | default `always-on` | `intermittent`: offline is normal (laptop), not an incident |
| `unlock` | object, see Disk unlock | | Leave it out for an unencrypted disk |
| `runsOn` | host id | | Only `vm`/`container`: the machine it runs on |
| `inside` | host id | | Physically built into another device |
| `power` | host id | | Power source: PDU/UPS, PoE switch, or the host a card sits in |
| `manages` | list of host ids | | Only `kvm`: machines it gives out-of-band access to |
| `sshHostKeys` | list of strings | | SSH host public keys, `"<type> <base64>"` without comment, see SSH host keys |
| `interfaces` | object: name → interface | | Network interfaces, see Interfaces |
| `note` | string | | Free text |

#### Kinds { #kinds }

| Kind | For |
|---|---|
| `server` | Physical machine running services |
| `hypervisor` | Physical machine running VMs/containers for others |

| Kind | For |
|---|---|
| `vm`, `container` | Virtual; the only kinds that may have `runsOn` |
| `router`, `switch`, `ap` | Network (`ap` = WLAN access point) |
| `pdu`, `ups` | Power |
| `kvm` | IP-KVM, external or as a card inside its host; the only kind that may have `manages` |
| `nas`, `workstation`, `laptop` | As named |
| `appliance` | Anything else with an address: Apple TV, printer, … |

Whether a host is virtual follows from `kind` (`vm`, `container`) – not from `runsOn`. A rented VM is a `vm` without `runsOn`: the provider's hypervisor is not in the inventory.

#### networks

A layer-2 network with its IPv4 prefix: the home LAN, a VLAN in the hackerspace.

| Field | Type | | Meaning |
|---|---|---|---|
| `name` | string | required | Display name (the id is the technical key) |
| `site` | site id | required | Where the network is |
| `vlan` | integer 1–4094 | | 802.1Q VLAN id; leave out for an untagged network. Unique per site – another site may use the same id |
| `ipv4` | prefix, e.g. `192.0.2.0/24` | | Without host bits |
| `ipv6` | prefix, e.g. `2001:db8:1::/64` | | Only when it is fixed (ULA, static GUA) – not an ISP prefix that changes |
| `dhcpServer` | host id | | Serves DHCP for IPv4; must be at the site and have an interface in this network |
| `dhcpPool` | `{ "start": ip, "end": ip }` | | Dynamic range; inside `ipv4`, start ≤ end. Needs `ipv4` and `dhcpServer` |
| `note` | string | | Free text |

A network without any interface is allowed – it can be entered before its devices are.

### Interfaces { #interfaces }

`hosts.<id>.interfaces` maps interface names to their data. The name is what the OS calls it (`eno1`, `wlp2s0f0`, `en0`); for an appliance any label (`lan`, `wan`). A VLAN interface on a trunk (`eno1.20`) is an entry of its own; its VLAN id comes from its network.

```
"interfaces": {
  "eno1":    { "mac": "52:54:00:12:34:56", "medium": "ethernet", "linkLocal": "eui64",
               "network": "lan", "ipv4": { "mode": "reserved", "address": "192.0.2.10" } },
  "eno1.20": { "network": "iot", "linkLocal": "fe80::1234", "ipv4": { "mode": "dhcp" } }
}
```

| Field | Type | | Meaning |
|---|---|---|---|
| `mac` | `aa:bb:cc:dd:ee:ff`, lower case | | Hardware address. Unique lab-wide, no multicast address |
| `medium` | `ethernet` `wifi` | | Physical medium |

| Field | Type | | Meaning |
|---|---|---|---|
| `linkLocal` | `eui64` `stable-privacy` `random` or an `fe80::` address | | How the IPv6 link-local address is formed |
| `network` | network id | | Network it is in; at the same site as the host |
| `ipv4` | `{ "mode": …, "address": … }` | | IPv4 addressing, needs `network` |
| `note` | string | | Free text |

**Link-local.** Only with EUI-64 does the address follow from the MAC (`fe80::` + MAC with the U/L bit flipped and `ff:fe` in the middle); the derived `ipv6LinkLocal` is computed then. Many systems use `stable-privacy` instead (RFC 7217: stable, but not computable), macOS and tvOS do not derive it from the MAC either. When the address is stable and known, enter it directly (`"fe80::…"`); `just inventory-collect` does exactly that when it does not match EUI-64. `eui64` needs a `mac`; with a *locally administered* Wi-Fi MAC (private Wi-Fi address) it is an error, because that MAC – and with it the address – is usually random. A locally administered Ethernet MAC is fine (QEMU's `52:54:00:…` is one).

**IPv4 – `ipv4.mode`:**

| Mode | Meaning | Rules |
|---|---|---|
| `reserved` | Fixed DHCP lease by MAC (reservation on `dhcpServer`) | `address` and `mac` required; the network needs a `dhcpServer`. MAC not looked up yet: the placeholder `aa:bb:cc:dd:ee:ff` - never passed on (`mac` null in the derived view, not in `reservations`), listed under `pending` |
| `static` | Configured on the device itself | `address` required; outside `dhcpPool` |
| `dhcp` | Dynamic | no `address` |

Every `address` lies inside the network's `ipv4` prefix, is neither network nor broadcast address (except /31, /32) and occurs once per network. A network without `ipv4` cannot take addresses.

**IPv6** is in here only as link-local and as a fixed prefix per network: DHCPv6 reservations hang on the DUID, not the MAC, and an ISP prefix (SLAAC) changes.

## SSH host keys { #ssh-host-keys }

`sshHostKeys` holds a device's public host keys – for managed and unmanaged devices alike – as `"<type> <base64>"` without the comment: `ssh-ed25519`, `ecdsa-sha2-nistp256/384/521` or `ssh-rsa`. One ed25519 key is enough; others only when the device has no ed25519 (Dropbear, KVMs). Checked: the format, that the key data matches its type (the base64 starts with the encoded type name), and that no key occurs twice – a cloned VM that kept its host key would show up that way.

Where they go:

| Consumer | How |
|---|---|
| Every NixOS host and zorlock | `programs.ssh.knownHosts` from `lib.inventory.knownHosts` (in `flake.nix`, the module `inventoryKnownHosts` in every host's list). Deploys and ursalia's unattended push to cuckoosegg need no trust-on-first-use |
| gusto (workstation) | `programs.ssh.knownHosts` like every host – `/etc/ssh/ssh_known_hosts` after the next `just mac-switch` |
| A workstation outside this repo | `just known-hosts` writes `~/.ssh/known_hosts.nixlab`; ssh reads it with `UserKnownHostsFile ~/.ssh/known_hosts ~/.ssh/known_hosts.nixlab` under `Host *` in `~/.ssh/config` |

| Consumer | How |
|---|---|
| `.sops.yaml` | `checks.inventory-sops`: for every managed host, `ssh-to-age` of its ed25519 key must give the age recipient `&<host>` in `.sops.yaml`. Catches a host key that changed (reinstall, rotation) without `sops updatekeys`, and a mistyped key. Hosts without a key yet are skipped |

Names per key: the host's id, for managed hosts `<host>.peer.homelab.example` (the mesh name), and every fixed IPv4 address of its interfaces (`static`/`reserved`). One entry per key, named `<host>` or `<host>-<n>` with several keys.

## Collecting from the devices

`just inventory-collect` fetches what is tedious to type:

- **Managed hosts**, over the mesh (root, zorlock as `admin`):
  - host keys via `ssh-keyscan` (only the ed25519 one when there is one);
  - per interface name, MAC, `ethernet`/`wifi` and link-local – `eui64` when the address matches the MAC, otherwise the address itself. On Linux every interface with a device behind it (hardware or virtio; no `lo`, mesh, tunnels, bridges, veths), on macOS the hardware ports without Thunderbolt;
  - for hardware that is not virtual: `vendor`, `model`, `productId`, `serial`. On Linux from `/sys/class/dmi/id` (Lenovo: model from `product_version`, machine type from `product_name`; everybody else: model from `product_name`), on macOS from `system_profiler` (model as "Mac mini (M4, 16 GB)", the model identifier as `productId`). Placeholders like "To be filled by O.E.M." are dropped.
- **Unmanaged devices**: only host keys, at their first fixed IPv4 address – enter `network`/`ipv4` first.

| Call | Effect |
|---|---|
| `just inventory-collect` | All devices; **adds** only what is missing, a differing value is printed and kept |
| `just inventory-collect ursalia beryl` | Only these devices |
| `just inventory-collect --force ursalia` | Takes the device's value where it differs, and prints each value it overwrote |

A changed host key should be a decision, hence `--force` – best with the host named. Even `--force` never touches what the device does not report: `network`, `ipv4`, notes, and interfaces it no longer has (remove those by hand).

## Relations

| Field | Meaning | May point at | Derived reverse |
|---|---|---|---|
| `runsOn` | Virtualisation | `vm` → `hypervisor`; `container` → `hypervisor`, `server`, `nas`, `vm` | `guests` |
| `inside` | Physical installation (PCIe card, module) | any host | `contains` |
| `power` | Power supply | any host | `powers` |
| `manages` | Out-of-band access (on the KVM) | any host but itself | `managedBy` |
| `unlock.tang` | Tang server for NBDE | any host but itself | `tangClients` |
| `interfaces.*.network` | Network membership | a network at the host's site | `networks.<id>.interfaces` |
| `networks.dhcpServer` | DHCP for a network | a host with an interface in it | – |

`runsOn`, `inside` and `power` together form the dependency graph: what a host cannot work without. It must be free of cycles; the derived `upstream` of a host is its transitive closure. `runsOn` and `inside` exclude each other.

`inside` says nothing about power: whether a PCIe KVM card lives off its host's slot (standby power) or off PoE/USB is what `power` says. A card that only draws from the slot gets `"power"` set to its host. Then "host and KVM both gone" points at power or network, "host gone, KVM still there" at the OS – and the KVM is the way in.

## Disk unlock

```
"unlock": { "methods": ["tpm", "nbde"], "threshold": 1, "tang": ["nas"], "fallback": "remote-
        ssh" }
```

| Field | Type | | Meaning |
|---|---|---|---|
| `methods` | list of `tpm`, `nbde` | required (may be empty) | Automatic methods. Empty: only a human can unlock |
| `threshold` | integer ≥ 1 | default 1 | How many methods must succeed – the Clevis `sss` threshold. 1 = any, 2 = TPM **and** Tang |
| `tang` | list of host ids | required with `nbde` | Tang servers; any one of them is enough |
| `fallback` | `remote-ssh` `console` | | What a human does when the methods fail: SSH (initrd SSH, FileVault over SSH) or console/KVM |

| Case | `unlock` |
|---|---|
| Unencrypted | leave out |
| TPM, passphrase at the console as fallback | `{"methods": ["tpm"], "fallback": "console"}` |
| Only a human, over SSH (e.g. FileVault) | `{"methods": [], "fallback": "remote-ssh"}` |
| TPM **or** Tang | `{"methods": ["tpm", "nbde"], "tang": ["a"]}` |
| TPM **and** Tang (disk only unlocks in its own network) | `{"methods": ["tpm", "nbde"], "threshold": 2, "tang": ["a"]}` |
| Only Tang, two servers | `{"methods": ["nbde"], "tang": ["a", "b"], "fallback": "console"}` |

NBDE is a dependency of its own: a host that **needs** Tang (`nbde` without `tpm`, or `threshold` 2) only comes back once one of its Tang servers is up unattended itself. The validator follows these chains. Is there a host that would never come up without a human – two hosts that are each other's only Tang server, or a Tang server that itself waits at a passphrase prompt – that is an error. The derived `unattendedBoot` says per host whether it comes back after a reboot or power cut on its own.

Not checked: whether a Tang server actually runs Tang. That lives in its config, not here.

## Validation

`nix flake check` (or just the inventory part: `just inventory-test`) validates in three stages. Each stage only runs when the one before found nothing, so a typo does not turn into twenty follow-up errors. Every error names the path:

**1. Structure** – top level first, on its own, then every entry:

- only `$schema`, `owners`, `uplinks`, `sites`, `networks`, `hosts` at the top, all five sections present
- valid ids and interface names, no unknown fields (catches typos), required fields present
- types, enums, minimums/maximums, formats (MAC, IPv4 address and prefix, IPv6 prefix, link-local, SSH key), no duplicates in lists
- references point at an existing entry of the right section

```
hosts.vps.critcality: unknown field (known: availability, criticality, inside, interfaces, kind,
managed, manages, model, note, owner, power, productId, runsOn, serial, site, sshHostKeys, unlock,
vendor)
hosts.nas.interfaces.eno1.mac: "52-54-00-12-34-56" is not a lower-case MAC address like
aa:bb:cc:dd:ee:ff
hosts.app.runsOn: "pve2" is not in hosts
uplinks.fiber-home.asn: expected a whole number, got "AS64496"
```

**2. Rules between entries:**

- `runsOn` only for `vm`/`container` and only on an allowed kind; not together with `inside`
- `manages` only for `kvm`; nothing points at itself
- `runsOn`/`inside`/`power` without cycles
- `site` present or inheritable; an explicit `site` matches the inherited one
- every site has an uplink; no uplink, site or owner without use; IPv6 details of an uplink only with `ipv6`
- `unlock`: at least a method or a fallback; `threshold` ≤ number of methods; `tang` exactly when `nbde`
- `managed` agrees with `flake.nix`: every `nixosConfigurations`/`darwinConfigurations` entry is in here as managed, and every managed entry has a configuration
- hardware: `vendor`/`productId`/`serial` not on `vm`/`container`; no serial twice
- SSH keys: key data matches the type, no key on two devices
- interfaces: network at the host's site; `ipv4` needs a network, `address` according to the mode, inside the prefix, not network/broadcast, static outside the pool, unique per network; `reserved` needs `mac` and a `dhcpServer`; MAC unique and not multicast; `eui64` needs a MAC and no private Wi-Fi MAC
- networks: prefix without host bits; pool inside the prefix, start ≤ end, only with `ipv4` and `dhcpServer`; `dhcpServer` at the site and with an interface in the network; VLAN id unique per site

```
hosts.app.runsOn: a vm cannot run on nas (kind nas; allowed: hypervisor)
hosts: runsOn/inside/power form a cycle through pve, ups1
hosts.laptop.managed: true, but flake.nix has no configuration of that name
hosts.router.interfaces.lan.ipv4.address: 192.0.2.150 is static but inside the DHCP pool of
network lan
networks.lan.vlan: 20 is also used by networks.iot at site home
hosts.router.sshHostKeys[0]: same key as hosts.nas.sshHostKeys[0] - a cloned machine that kept its
host key?
```

**3. Unlock chains** – see Disk unlock:

```
hosts.nas.unlock: never unlocks unattended - none of its Tang servers (vps) comes up without a
human
```

**Separately**, as its own check because it needs `ssh-to-age`: `checks.inventory-sops` (see SSH host keys).

```
hosts.cuckoosegg: host key gives age1z9rt…, .sops.yaml has age1z8uz… - host key changed without
sops updatekeys, or a wrong key in inventory.json
```

The schema in `inventory.schema.json` only covers stage 1 (without references) and is there for the editor; the validation in `inventory.nix` is what counts.

## Derived view

`lib.inventory.inventory` (on the console: `just inventory`) is the validated inventory with everything derived. Every section gets its `id`; additionally:

| Section | Derived |
|---|---|
| `owners.<id>` | `hosts` – the hosts it administers |
| `uplinks.<id>` | `sites` – the sites on this line |
| `sites.<id>` | `hosts` – everything there, inherited sites included |
| `hosts.<id>` | `site` (effective, inherited along `runsOn`/`inside`), `virtual`, `uplinks` (of its site, primary first), `guests`, `contains`, `powers`, `managedBy`, `tangClients`, `upstream` (transitive, nearest first), `unattendedBoot` |
| `hosts.<id>.interfaces.<name>` | `name`, `vlan` (from its network), `ipv6LinkLocal` (computed for `eui64`, as given for a fixed address, otherwise `null`) |
| `pending` | every placeholder MAC still to replace; `just inventory-test` prints the list |
| `networks.<id>` | `interfaces` – every interface in it (`host`, `interface`, `mac`, `address`, `mode`); `reservations` – what `dhcpServer` has to hand out by MAC (`host`, `interface`, `mac`, `address`), sorted by address |

Besides `inventory`: `lib.inventory.knownHosts` (the `programs.ssh.knownHosts` value) and `lib.inventory.knownHostsFile` (the same as known_hosts text).

Optional fields are always present in the derived view: defaults filled in (`criticality`, `availability`, `unlock.threshold`), otherwise `null` or `[]`. A consumer never has to write `or`.

If the inventory is invalid, `inventory` throws with **all** errors at once; `lib.inventory.errors` is the list itself (empty when valid).

## Examples

The examples are the test inventory from `tests/inventory.nix`, written as JSON. It covers every field and relation.

**Rented VM** – virtual, the provider's hypervisor is not in the inventory:

```
"vps": { "kind": "vm", "site": "dc", "owner": "lab", "managed": true, "criticality": "core" }
```

**VM on a hypervisor of others, which hangs on a UPS** – `site` is inherited from `pve`:

```
"pve":  { "kind": "hypervisor", "site": "home", "owner": "club", "managed": false, "power":
          "ups1" },
"ups1": { "kind": "ups", "site": "home", "owner": "lab", "managed": false },
"app":  { "kind": "vm", "runsOn": "pve", "owner": "lab", "managed": true,
          "unlock": { "methods": ["nbde"], "tang": ["nas"] } }
```

Derived for `app`: `site` = `home`, `upstream` = `["pve", "ups1"]`, `unattendedBoot` = `true` (via the chain app → nas → vps).

**NAS with TPM and Tang required together, plus a PCIe KVM card inside it:**

```
"nas": { "kind": "nas", "vendor": "Example Corp", "model": "Example NAS", "productId": "EX-1000",
         "serial": "NAS0001", "site": "home", "owner": "lab", "managed": true,
         "unlock": { "methods": ["tpm", "nbde"], "threshold": 2, "tang": ["vps"] } },
"kvmcard": { "kind": "kvm", "inside": "nas", "power": "nas", "manages": ["nas"],
             "owner": "lab", "managed": false }
```

Derived for `nas`: `contains`, `powers` and `managedBy` = `["kvmcard"]`, `tangClients` = `["app"]`.

**Laptop** – offline is normal, only a human unlocks it:

```
"laptop": { "kind": "laptop", "site": "home", "owner": "lab", "managed": true,
            "availability": "intermittent", "unlock": { "methods": [], "fallback": "console" } }
```

**Networks, a reservation and a router** – the router hands out `nas`'s address by MAC; its own address is static, outside the pool:

```
"networks": {
  "lan": { "name": "Home LAN", "site": "home", "ipv4": "192.0.2.0/24", "ipv6": "2001:db8:1::/64",
           "dhcpServer": "router", "dhcpPool": { "start": "192.0.2.100", "end": "192.0.2.199" } },
  "iot": { "name": "IoT", "site": "home", "vlan": 20, "ipv4": "198.51.100.0/24" }
},
"router": { "kind": "router", "site": "home", "owner": "lab", "managed": false,
            "sshHostKeys": ["ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI…"],
            "interfaces": { "lan": { "network": "lan", "ipv4": { "mode": "static", "address": "192
        .0.2.1" } } } }
```

Derived: `networks.lan.reservations` = `nas/eno1` (192.0.2.10) and `app/ens18` (192.0.2.11); `nas.interfaces.eno1.ipv6LinkLocal` = `fe80::5054:ff:fe12:3456`; known_hosts names for `nas` = `nas`, `nas.peer.example`, `192.0.2.10`.

**Site with primary line and LTE backup:**

```
"uplinks": {
  "fiber-home": { "isp": "Example ISP", "asn": 64496, "medium": "fiber",
                  "bandwidth": { "down": 1000, "up": 500 }, "ipv4": "cgnat", "ipv6": true },
  "lte-backup": { "medium": "lte" }
},
"sites": {
  "home": { "city": "Exampletown", "kind": "residential", "uplinks": ["fiber-home", "lte-
        backup"] }
}
```

**Reading it elsewhere.** In Nix (another flake output, a module via `specialArgs`):

```
self.lib.inventory.inventory.hosts.app.upstream   # [ "pve" "ups1" ]
```

In OpenTofu – the raw data, without derived values and without validation (that stays with `nix flake check`):

```
locals {
  inventory = jsondecode(file("${path.module}/../../inventory.json"))
  servers   = [for id, h in local.inventory.hosts : id if h.managed]
}
```

## Tests

`tests/inventory.nix` holds a complete, fictional inventory (the examples above) and 72 cases. Each case changes exactly one thing and states the expected errors – exact wording and order – or, for valid variants, values of the derived view. A case also fails if a broken inventory does not make `.inventory` throw.

| Group | Covers |
|---|---|
| valid | Base inventory with all derived values; container on a VM; Tang loop broken by `threshold 1`; flake check switched off |
| uplinks | IPv6 prefix length out of range, IPv6 details without IPv6 |
| structure | Unknown/missing top-level key, typo in a field, enum, required field, type, minimum, dangling and wrong-section references, invalid id, duplicates |
| rules | `runsOn` on non-virtual and on a wrong kind, `runsOn` + `inside`, `manages` outside a KVM and on itself, self reference, cycle, missing/non-inheritable/contradicting `site`, site without uplink, unused entries, inventory ≠ flake |
| unlock | No way to unlock, `threshold` too high or without methods, `nbde` without `tang` and the reverse, own Tang server, Tang loop, Tang server waiting for a human |
| hardware | `vendor`/`serial` on a VM, `model` on a VM allowed, serial twice; derived hardware fields and defaults |
| SSH keys | Wrong format, with comment, key data of another type, same key on two hosts; derived known_hosts entries and file |
| interfaces | MAC format, interface name, link-local value, MAC twice, multicast MAC, `eui64` from a private Wi-Fi MAC and without MAC, network at another site, `ipv4` without network, address per mode, reserved without MAC or DHCP server, address outside/broadcast/in pool/without prefix/twice; derived link-local (EUI-64 of a QEMU MAC, fixed, random) and VLAN |
| networks | Prefix format, VLAN range and duplicate per site (same id at another site allowed), host bits, pool outside/backwards/without server, DHCP server without interface or at another site; derived reservations |

```
just inventory-test
```

builds `checks.<system>.inventory` (the real file is valid), `inventory-tests` (all cases pass), `inventory-schema` (schema fresh) and `inventory-sops` (host keys ↔ `.sops.yaml`). On failure the message lists each failing case with "want" and "got".

**A new rule** gets a case in the same commit: change one thing in `base` with `set`/`del`, paste the exact message. **Changed wording** of an error fails the tests – adjust case and this page together.

## Workflows

| What | Steps |
|---|---|
| New host / device | Add an entry to `inventory.json` (for a flake host `"managed": true` – otherwise `nix flake check` fails anyway), `just inventory-collect`, `network`/`ipv4` by hand, `just inventory-test` |
| Host key changed (reinstall) | `just inventory-collect --force <host>`, `.sops.yaml` + `sops updatekeys` (see Rotate keys), deploy, `just known-hosts` |
| New DHCP reservation | Interface with `mac`, `network`, `ipv4.mode: reserved` + `address`; the list for the router: `networks.<id>.reservations` in `just inventory` |
| Fill in a line's details | Fields in `uplinks.<id>`, nothing else to do |

| What | Steps |
|---|---|
| New field or kind | `spec` in `inventory.nix`, then `just inventory-schema`, a test case in `tests/inventory.nix`, this page (de/en) |
| Look at it | `just inventory` |

## Services

Services are named by function, not by product. There are three kinds of reachability: public directly, public behind the NetBird reverse proxy with SSO, or mesh only.

| Name (`.homelab.example`) | App | Host | Reachable |
|---|---|---|---|
| `auth` | Pocket ID | zummi | public (nginx) |
| `mesh` | NetBird management/ dashboard | grammi | public (Traefik terminates TLS) |
| `s3.aldrin` | Garage S3 API | aldrin | public (nginx) |
| `rezepte` | Mealie | grammi | reverse proxy + SSO |
| `grist` | Grist | grammi | reverse proxy + SSO |
| `logs` | logchef | grammi | reverse proxy + SSO |
| `metrics` | Grafana | grammi | reverse proxy + SSO |
| `vm` / `vl` | VictoriaMetrics / VictoriaLogs UI | grammi | reverse proxy + SSO (via Traefik entrypoint `apps`, port 8090) |
| `s3-admin` | garage-ui | aldrin | mesh only |
| `dokumente` | Paperless + paperless-gpt (`/paperless-gpt/`) | ursalia | in the mesh directly; from outside, through the reverse proxy, share links via PIN and the app via SSO (why) |
| `llm` | Ollama (behind Caddy) | zorlock | mesh only |
| `mqtt` | Mosquitto (Home Assistant add-on) | Home Assistant (home LAN, NetBird peer) | port 8883, MQTT over TLS: at home straight to HA (home router (OPNsense) answers the name locally), from outside through the reverse proxy as TLS passthrough to the peer, no SSO (why) |
| `acme` | dns01proxy (DNS-01 for Home Assistant) | grammi | mesh only, port 8446; HA resolves it through the home router (OPNsense) (why) |
| `doku` | This documentation (private version) | aldrin + ursalia | in the mesh admin devices only (why), one A record per host; from outside a redirect to the public docs through the reverse proxy |

**Reverse proxy:** Traefik on grammi owns port 443. It terminates `mesh.homelab.example` itself and hands everything else via TCP passthrough with PROXY protocol v2 to `netbird-proxy`, which does TLS and ACME (tls-alpn-01) on its own. Login at the proxy is SSO against Pocket ID, limited to the NetBird groups of the respective app; `dokumente` additionally with a PIN (share links only).

**Mesh-only names:** Every name is its own NetBird DNS zone with one A record pointing at the mesh address (192.0.2.x). Public DNS does not know them; the certificate comes via DNS-01 through deSEC.

**Across all Linux hosts:** vmagent ships metrics, Vector ships journald and nginx logs to grammi. CrowdSec reads the nginx/Traefik logs on the public hosts. restic backups go to Garage buckets on aldrin, for Paperless additionally append-only to BorgBase.

# Modules

The 17 modules under `modules/` carry the configuration shared by all hosts; every file explains in its header comment why it is the way it is. The public version contains them anonymised under `modules/` – as a reference, not directly importable: some read `hosts/<host>/*.yaml` and assume sops-nix. Four have options of their own.

| Module | Purpose | Options / notes |
|---|---|---|
| auto-upgrade-git.nix | Nightly `nixos-rebuild switch`, commits and pushes the new `flake.lock` back to the repo | `services.autoUpgradeGit`: `enable`, `flakeDir`, `flakeAttr`, `updateInputs`, `dates`, `allowReboot`, `rebootWindow`, `deployKeySecret`, `gitAuthorEmail` |
| log-shipping.nix | journald and nginx logs to VictoriaLogs via Vector | `services.logShipping`: `enable`, `victoriaLogsUrl` |
| metrics-shipping.nix | vmagent scrapes the local exporters and remote-writes to VictoriaMetrics | `services.metricsShipping`: `enable`, `remoteWriteUrl`, `exporterPorts` |
| deploy.nix | The lab's policy for auto-upgrade-git: GitHub deploy key, pinned host key, reboot window 04:00–06:00 | kept apart from the mechanism so that stays reusable |
| acme.nix | ACME defaults for every TLS host: DNS-01 against deSEC | token from `hosts/<host>/desec.yaml` (from Tofu) |
| netbird.nix | NetBird client in the mesh, registration via setup key, local metrics | setup key from `hosts/<host>/netbird.yaml` (from Tofu) |
| tailscale.nix | Shared Tailscale agent | `tailscale serve` stays in the host config |
| ssh.nix | Who may log in from where: root only via mesh or Tailscale | keys stay per host |
| firewall.nix | Only 22/80/443 public, tailnet as trusted interface | |
| crowdsec.nix | CrowdSec engine and firewall bouncer, whitelist for the lab hosts | workarounds for bugs in the nixpkgs modules, documented in the header |
| networking.nix | ifstate skeleton, systemd-resolved with DNS over TLS, no fallback resolvers | addresses and routes stay in the host config |
| metrics.nix | Prometheus exporters per host | protected by the firewall instead of bind addresses |
| logging.nix | Bounds journald, which otherwise grows up to 4 GB | |
| qemu-guest.nix | QEMU guest agent with an RPC deny list: snapshots yes, shell and file access over the channel no | |
| nix.nix | Nix daemon, garbage collection, store optimisation | |
| docs-site.nix | The private docs site (flake package `docs-site`) under `doku` on the mesh | `services.docsSite`: `enable`, `package`, `fqdn`, `allow` (default: the whole mesh; in the lab only the admin devices from `admin-peers.json`); on aldrin and ursalia |
| base.nix | Locale, console, time and tools for debugging | |

# just recipes

| Recipe | Purpose |
|---|---|
| `just check <host>` | `git pull` + `nixos-rebuild dry-activate` |
| `just secrets <host>` / `pull-secrets <host>` | Edit host secrets on the host and fetch them back |
| `just topology` | Regenerate the topology dashboard (otherwise `nix flake check` fails) |
| `just inventory` | Show the validated inventory with everything derived (Inventory) |
| `just inventory-test` | Check `inventory.json`, run the validator tests, check the schema is fresh |
| `just inventory-schema` | Regenerate `inventory.schema.json` after a format change |
| `just inventory-collect [--force] [host …]` | Fetch SSH host keys, interfaces (MAC, medium, link-local) and hardware (vendor, model, product id, serial) from the devices into `inventory.json` – fills gaps and reports differences; `--force` takes the device values |
| `just known-hosts` | Write `~/.ssh/known_hosts.nixlab` from the inventory – for a workstation outside this repo; gusto gets the keys system-wide |
| `just netbird-plan` / `netbird-apply` | Change the NetBird account; apply also rewrites `expected-peers.json` (commit it, deploy grammi) and `admin-peers.json` (commit it, deploy aldrin and ursalia) |
| `just netbird-proxy-certs` | Get proxy services without a certificate one and bring the dashboard up to date (NetBird #7118); runs at the end of `netbird-apply` |
| `just dokumente-share-pin` / `dokumente-share-pin-show` | New random PIN for Paperless share links from outside (then `netbird-apply`), or show the current one |
| `just netbird-key <host>` | New setup key for a host, 24 h to deploy |
| `just netbird-import` | Restore lost Tofu state from the account |
| `just netbird-renew-token` | Yearly renewal of the service user token |
| `just github-key <host>` | New GitHub deploy key for a host (key via sops, registration via Tofu) |
| `just github-plan` / `github-apply` | Change the repositories and deploy keys on GitHub |
| `just desec-key <host>` / `desec-verify` | Rotate a deSEC token, or check the scopes from every host |
| `just nanokvm-mcp-key` | Store the NanoKVM Go+ MCP API key encrypted for ursalia (then commit, deploy ursalia) |
| `just kvm-bridge setup\|start\|stop\|status\|log` | Claude Code bridge to the NanoKVM Go+ on ursalia: set up once, start, stop, state, log |
| `just mac-diff [host]` / `mac-switch [host]` | Macs (default gusto, or zorlock over SSH): show what a switch would change (nvd, Homebrew removals/installs), or switch with a summary – see macOS hosts |
| `just mac-upgrade` | On gusto: update the Homebrew pins, switch, `brew upgrade`, NetBird version next to the mesh pin; commit `flake.lock` |
| `just gc <host>` | Delete store generations older than 7 days |
| `just docs` | Build this documentation: private PDFs (de/en) and the anonymised public version (README, PDFs, site sources, modules) in `public-docs/` (gitignored clone of `nixlab-docs`), committed and pushed there; GitHub builds the site from it |
| `just docs-serve` | Build the private site with real values locally and serve it at `http://127.0.0.1:8000` |

Recipes that need sops get the workstation's age key from 1Password through `SOPS_AGE_KEY_CMD`. On gusto home-manager sets it and `op` unlocks through the 1Password app (Touch ID); sops, tofu and age come from the devShell (direnv in the checkout). Without that setup:

```
eval $(op signin) && export SOPS_AGE_KEY_CMD='op read "op://<vault>/<item>/password"' && just
        netbird-plan
```

## Troubleshooting

- After a `nixos-rebuild`, `systemctl --failed` says nothing about the past – the switch clears failed states. Use `journalctl -u <unit>` instead.
- NetBird down, Tailscale still there: put `NIXLAB_DOMAIN=example-tailnet.ts.net` in front of the recipe.
- Tofu recipes refuse to run against an empty state, so no second `servers` group or set of tokens gets created.
- WLAN on ursalia: `iw dev wlp2s0f0 link` shows band, signal and PHY mode. The router's 5 GHz radio is on a DFS channel with a 300 TU beacon interval, so a single scan can miss it; band, signal, rates and PHY mode also go to VictoriaMetrics (node_exporter `wifi` collector plus `wifi_link_*`), Grafana dashboard "Homelab / WLAN".
- After a network change NetBird can keep a dead P2P path for minutes while still showing `Connected` (seen on ursalia, 2026-10-19). Fix: `systemctl restart netbird-mesh` on the host, via Tailscale if the mesh is down.
- A proxy service hangs at "Issuing certificate..." in the NetBird dashboard or fails the TLS handshake from outside ("internal error"): `just netbird-proxy-certs` – requests the names without a certificate from outside and restarts `netbird-proxy` on grammi (NetBird discussion #7118). Also runs at the end of `just netbird-apply`.

## Conventions

- One git worktree per topic (`.worktrees/<topic>`), removed and its branch deleted when done. Tofu state always stays in the main checkout.
- Plan first, implement after approval.
- Subdomains by function (`rezepte`, not `mealie`).
- New web services on grammi use PostgreSQL instead of SQLite.
- Dual stack wherever possible.
- Code comments explain the *why*, including the mistakes that led to a decision, with a date.
- Code comments, commit messages and changelogs in English; the project documentation exists in German and English.
- The docs site loads nothing from third parties: the fonts and glightbox ship with the site (`docs/assets.nix`) instead of coming from Google Fonts or unpkg.com. Why: GDPR, and private docs that work without internet access.

## Open items

☐ Move the NetBird mesh to dual stack (IPv6 overlay)
☐ Upstream PRs to the NetBird provider (#247 `direct_upstream`/cluster targets, #248 `crowdsec_mode`); then switch the proxy targets on grammi to `localhost`
☐ Remove Tailscale from the servers once nothing depends on it any more
☐ Grafana alerting (contact point still missing) for SMART, fwupd, Paperless, NetBird expected ≠ known, ursalia's WLAN (not associated, stuck on 2.4 GHz, weak signal)
☐ Update notifications for the inputs pinned by hand (NetBird, Pocket ID)
☐ Declarative root passwords via sops
☐ Keep the host list in one place instead of `flake.nix`, the Tofu modules and `justfile`
☐ gusto: after the first switch check whether the NetBird app updates itself (or follows the dashboard's auto-update) and switch that off, so the version comes from `netbirdio-tap` only
☐ gusto: default git address `admin@homelab.example` instead of `mail@admin.org`, together with a new layout of `~/Projects`
☐ Inventory: a field for a device's DNS name (arkkvm answers as `arkkvm01`, today only in its note), and public addresses for the managed hosts – then `hosts/gusto/ssh.nix` derives its fallback names from them